Vector 3D: Questions and Full Solutions

Question 01

English

Let $A(3,1,0)$, $B(1,3,2)$ and $C(-1,1,4)$ be three points. Let L denote the set of all points P in space such that $PA = PB = PC$. Then the point at which L meets the plane $x + y + z = 5$ is

(A) $(1,0,2)$
(B) $(3,0,2)$
(C) $(2,0,3)$
(D) $(2,1,2)$

Only one option is correct.

हिन्दी

माना $A(3,1,0)$, $B(1,3,2)$ तथा $C(-1,1,4)$ तीन बिन्दु हैं। माना L त्रिविम समष्टि के उन सभी बिन्दुओं P का समुच्चय है जिनके लिए $PA = PB = PC$ है। तब वह बिन्दु, जिस पर L समतल $x + y + z = 5$ से मिलता है, है

(A) $(1,0,2)$
(B) $(3,0,2)$
(C) $(2,0,3)$
(D) $(2,1,2)$

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Solution

Let $P(x,y,z)$ be a point of L. The condition $PA = PB = PC$ is equivalent to the two equations $PA^2 = PB^2$ and $PA^2 = PC^2$.

$PA^2 - PB^2 = \left[(x-3)^2 + (y-1)^2 + z^2\right] - \left[(x-1)^2 + (y-3)^2 + (z-2)^2\right] = -4x + 4y + 4z - 4.$

So $PA^2 = PB^2$ gives the plane $x - y - z + 1 = 0$, the perpendicular-bisector plane of AB.

$PA^2 - PC^2 = \left[(x-3)^2 + (y-1)^2 + z^2\right] - \left[(x+1)^2 + (y-1)^2 + (z-4)^2\right] = -8x + 8z - 8.$

So $PA^2 = PC^2$ gives the plane $x - z + 1 = 0$, the perpendicular-bisector plane of AC.

The two planes are not parallel (normals $(1,-1,-1)$ and $(1,0,-1)$), so L is their line of intersection. From $z = x + 1$ and $x - y - z + 1 = 0$ we get $y = 0$.

Hence $L: \dfrac{x}{1} = \dfrac{y}{0} = \dfrac{z-1}{1}$, that is, the points $(t, 0, t+1)$, $t \in \mathbb{R}$. Its direction $(1,0,1)$ is parallel to $\overrightarrow{AB} \times \overrightarrow{AC}$, so L is the normal to the plane ABC through the circumcentre of the triangle.

Substituting $(t,0,t+1)$ in $x + y + z = 5$: $2t + 1 = 5 \Rightarrow t = 2$, giving the point $(2,0,3)$.

Check: $PA^2 = 1 + 1 + 9 = 11$, $PB^2 = 1 + 9 + 1 = 11$, $PC^2 = 9 + 1 + 1 = 11$.

The other options fail: $(1,0,2)$ is the circumcentre of triangle ABC but not on the plane, $(3,0,2)$ and $(2,1,2)$ lie on the plane but are not equidistant from A, B, C.

Correct Option: (C)

Question 02

English

Let ℓ be the line of intersection of the planes $P_1 : x + 2y - z = 3$ and $P_2 : x - z = 1$, and let M be the line $\dfrac{x-1}{1} = \dfrac{y-1}{2} = \dfrac{z+3}{2}$. Among all the planes containing ℓ, exactly one plane, say Π_1, has no point in common with M. Let Π_2 be the plane containing ℓ which is perpendicular to Π_1, and let R be the point at which M meets Π_2. Then the distance of R from the line ℓ is

(A) 2
(B) $\dfrac{3\sqrt{2}}{2}$
(C) $\dfrac{2\sqrt{6}}{3}$
(D) 4

Only one option is correct.

हिन्दी

माना ℓ समतलों $P_1 : x+2y-z = 3$ तथा $P_2 : x-z = 1$ की प्रतिच्छेदन रेखा है, तथा M रेखा $\dfrac{x-1}{1} = \dfrac{y-1}{2} = \dfrac{z+3}{2}$ है। ℓ को समाहित करने वाले सभी समतलों में से ठीक एक समतल, माना Π_1, का M के साथ कोई उभयनिष्ठ बिन्दु नहीं है। माना Π_2 वह समतल है जो ℓ को समाहित करता है तथा Π_1 के लम्बवत् है, तथा R वह बिन्दु है जिस पर M समतल Π_2 से मिलती है। तब रेखा ℓ से R की दूरी है

(A) 2
(B) $\dfrac{3\sqrt{2}}{2}$
(C) $\dfrac{2\sqrt{6}}{3}$
(D) 4

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Solution

Every plane containing ℓ, other than P_2, is $(x + 2y - z - 3) + \lambda(x - z - 1) = 0$ for some real λ, with normal $\vec{n}_\lambda = (1 + \lambda,\, 2,\, -1 - \lambda)$.

A general point of M is $(1 + t,\, 1 + 2t,\, -3 + 2t)$. Substituting in P_2: $(1 + t) - (-3 + 2t) - 1 = 3 - t$, which vanishes at $t = 3$, so P_2 meets M.

Substituting in the member λ: $(1 + t) + 2(1 + 2t) - (-3 + 2t) - 3 + \lambda\,[(1 + t) - (-3 + 2t) - 1] = (3 + 3\lambda) + (3 - \lambda)t$.

This has no root t exactly when the coefficient of t vanishes and the constant does not: $3 - \lambda = 0$ and $3 + 3\lambda \neq 0$. So $\lambda = 3$, and this is the only such plane.

$\Pi_1 : (x + 2y - z - 3) + 3(x - z - 1) = 4x + 2y - 4z - 6 = 0$, i.e. $2x + y - 2z = 3$, with normal $(2, 1, -2)$ of length 3.

Members perpendicular to Π_1: $(1 + \mu, 2, -1 - \mu) \cdot (2, 1, -2) = 2 + 2\mu + 2 + 2 + 2\mu = 6 + 4\mu = 0 \Rightarrow \mu = -\dfrac{3}{2}$.

$\Pi_2 : (x + 2y - z - 3) - \dfrac{3}{2}(x - z - 1) = 0$, i.e. $x - 4y - z + 3 = 0$.

Key idea: Π_1 and Π_2 are perpendicular planes meeting in ℓ. For a point R of Π_2, the perpendicular from R to Π_1 lies in Π_2 and meets Π_1 on ℓ, so it is also the perpendicular from R to ℓ. Hence the distance of R from ℓ equals the distance of R from Π_1.

Since M is parallel to Π_1, every point of M is at the same distance from Π_1. For $(1, 1, -3)$:

$\dfrac{|2+1+6-3|}{3} = 2.$

Check by direct computation: M meets Π_2 where $(1+t)-4(1+2t)-(-3+2t)+3 = 3-9t = 0$, so $t = \dfrac{1}{3}$ and $R = \left(\dfrac{4}{3}, \dfrac{5}{3}, -\dfrac{7}{3}\right)$. A point of ℓ is $(1,1,0)$ and its direction is $(1,0,1)$; $\overrightarrow{AR} = \left(\dfrac{1}{3}, \dfrac{2}{3}, -\dfrac{7}{3}\right)$, $\overrightarrow{AR} \times (1,0,1) = \left(\dfrac{2}{3}, -\dfrac{8}{3}, -\dfrac{2}{3}\right)$, so the distance is $\dfrac{\sqrt{72}/3}{\sqrt{2}} = 2$.

Correct Option: (A)

Question 03

English

Let Π_1 and Π_2 be the planes $x + 2y + 2z = 12$ and $2x - 2y + z + 6 = 0$, respectively. Let B be the image of the point $A(1,0,1)$ in the plane Π_1, and let C be the image of the point B in the plane Π_2. Then C is the point

(A) $(3,4,5)$
(B) $(-1,8,3)$
(C) $(1,-8,-3)$
(D) $(-3,4,-1)$

Only one option is correct.

हिन्दी

माना Π_1 तथा Π_2 क्रमशः समतल $x+2y+2z = 12$ तथा $2x-2y+z+6 = 0$ हैं। माना समतल Π_1 में बिन्दु $A(1,0,1)$ का प्रतिबिम्ब B है, तथा समतल Π_2 में बिन्दु B का प्रतिबिम्ब C है। तब C बिन्दु है

(A) $(3,4,5)$
(B) $(-1,8,3)$
(C) $(1,-8,-3)$
(D) $(-3,4,-1)$

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Solution

For the plane $ax + by + cz + d = 0$, the image (x,y,z) of (x_1,y_1,z_1) lies on the normal through the point, at the same distance on the other side: $\dfrac{x-x_1}{a} = \dfrac{y-y_1}{b} = \dfrac{z-z_1}{c} = \dfrac{-2(ax_1+by_1+cz_1+d)}{a^2+b^2+c^2}$.

Image of $A(1,0,1)$ in $\Pi_1 : x + 2y + 2z - 12 = 0$: $ax_1 + by_1 + cz_1 + d = 1 + 0 + 2 - 12 = -9$, $a^2+b^2+c^2 = 9$, ratio $= \dfrac{18}{9} = 2$.

So $B = (1+2\cdot 1,\ 0+2\cdot 2,\ 1+2\cdot 2) = (3,4,5)$. Check: the midpoint $(2,2,3)$ satisfies $2+4+6 = 12$.

Image of $B(3,4,5)$ in $\Pi_2 : 2x - 2y + z + 6 = 0$: $6 - 8 + 5 + 6 = 9$, $a^2+b^2+c^2 = 9$, ratio $= -2$.

So $C = (3 - 2\cdot 2,\ 4 - 2(-2),\ 5 - 2\cdot 1) = (-1,8,3)$. Check: the midpoint $(1,6,4)$ satisfies $2 - 12 + 4 + 6 = 0$.

Further check: the normals satisfy $(1,2,2)\cdot(2,-2,1) = 0$, so the planes are perpendicular and the two reflections together are a half-turn about their line of intersection. The midpoint of AC is $(0,4,2)$, which lies on Π_1 $(0+8+4 = 12)$ and on Π_2 $(0-8+2+6 = 0)$, as it must.

Options (A) and (D) are the single images of A in Π_1 and in Π_2; (C) has a sign error.

Correct Option: (B)

Question 04

English

Let $A(1,0,1)$, $B(-1,1,2)$ and $C(2,2,1)$ be three points, and let D be a point of the parabola $y = x^2$ lying in the plane $z = 0$. As D varies on this parabola, the least value of the volume of the tetrahedron $ABCD$ is

(A) 6
(B) $\frac{7}{6}$
(C) $\frac{1}{2}$
(D) 1

Only one option is correct.

हिन्दी

माना $A(1,0,1)$, $B(-1,1,2)$ तथा $C(2,2,1)$ तीन बिन्दु हैं, तथा D समतल $z = 0$ में स्थित परवलय $y = x^2$ का एक बिन्दु है। जब D इस परवलय पर गति करता है, तब चतुष्फलक $ABCD$ के आयतन का न्यूनतम मान है

(A) 6
(B) $\frac{7}{6}$
(C) $\frac{1}{2}$
(D) 1

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Solution

A point of the parabola $y = x^2$ in the plane $z = 0$ is $D = (t, t^2, 0)$, $t \in \mathbb{R}$.

Edges from A: $\overrightarrow{AB} = (-2,1,1)$, $\overrightarrow{AC} = (1,2,0)$, $\overrightarrow{AD} = (t-1,\ t^2,\ -1)$.

The volume of the tetrahedron is $V = \frac{1}{6}\left|\left[\overrightarrow{AB}\ \overrightarrow{AC}\ \overrightarrow{AD}\right]\right|$.

Expanding along the first row: $\begin{vmatrix} -2 & 1 & 1 \\ 1 & 2 & 0 \\ t-1 & t^2 & -1 \end{vmatrix} = -2(-2-0) - 1(-1-0) + 1\left(t^2 - 2(t-1)\right) =$ $4 + 1 + t^2 - 2t + 2 = t^2 - 2t + 7$.

Completing the square, $t^2 - 2t + 7 = (t-1)^2 + 6 \geq 6$, so the scalar triple product is always positive: D never lies in the plane ABC.

Hence $V(t) = \dfrac{(t-1)^2 + 6}{6}$, which is least when $t = 1$, that is, at $D = (1,1,0)$, and the least value is $\dfrac{6}{6} = 1$.

(Option (B) is the volume at $t = 0$; option (A) forgets the factor $\frac{1}{6}$.)

Correct Option: (D)

Question 05

English

In the plane $z = 0$, the lines $x - y + 1 = 0$, $x + 2y - 11 = 0$ and $2x - y - 7 = 0$ form a triangle ABC, where A is the point of intersection of the second and the third line, B that of the first and the third line, and C that of the first and the second line. A point D, with positive z-coordinate, lies on the line through the origin parallel to the vector $2\hat{i} - \hat{j} + 2\hat{k}$, and the volume of the tetrahedron $ABCD$ is 10. Then the perpendicular distance of the vertex A from the plane BCD is

(A) $\frac{4}{3}$
(B) 2
(C) $\frac{8}{3}$
(D) $\frac{4}{5}$

Only one option is correct.

हिन्दी

समतल $z = 0$ में रेखाएँ $x - y + 1 = 0$, $x + 2y - 11 = 0$ तथा $2x - y - 7 = 0$ एक त्रिभुज ABC बनाती हैं, जहाँ A दूसरी तथा तीसरी रेखा का प्रतिच्छेदन बिन्दु है, B पहली तथा तीसरी रेखा का, तथा C पहली तथा दूसरी रेखा का प्रतिच्छेदन बिन्दु है। धनात्मक z-निर्देशांक वाला एक बिन्दु D मूलबिन्दु से होकर सदिश $2\hat{i} - \hat{j} + 2\hat{k}$ के समान्तर रेखा पर स्थित है, तथा चतुष्फलक $ABCD$ का आयतन 10 है। तब समतल BCD से शीर्ष A की लम्बवत् दूरी है

(A) $\frac{4}{3}$
(B) 2
(C) $\frac{8}{3}$
(D) $\frac{4}{5}$

केवल एक विकल्प सही है।

Solution

Vertices in the plane $z = 0$: A from $x + 2y = 11$, $2x - y = 7$ gives $(5, 3)$; B from $x - y = -1$, $2x - y = 7$ gives $(8, 9)$; C from $x - y = -1$, $x + 2y = 11$ gives $(3, 4)$. So $A = (5, 3, 0)$, $B = (8, 9, 0)$, $C = (3, 4, 0)$.

Area of triangle ABC (coordinate formula): $\frac{1}{2}\left|\begin{vmatrix} 5 & 3 & 1 \\ 8 & 9 & 1 \\ 3 & 4 & 1 \end{vmatrix}\right| = \frac{1}{2}\,|5(9-4) - 3(8-3) + 1(32-27)| =$ $\frac{1}{2}|25 - 15 + 5| = \frac{15}{2}$.

D is on the line through O parallel to $(2, -1, 2)$: $D = (2k, -k, 2k)$ with $2k > 0$. Its height above the base plane $z = 0$ is $2k$.

Volume $= \frac{1}{3}\times$ base area $\times$ height: $\frac{1}{3} \cdot \frac{15}{2} \cdot 2k = 5k = 10 \Rightarrow k = 2$, so $D = (4, -2, 4)$.

Face BCD: $\overrightarrow{BC} = (-5, -5, 0)$, $\overrightarrow{BD} = (-4, -11, 4)$.

$\overrightarrow{BC} \times \overrightarrow{BD} = ((-5)(4) - 0,\ 0 - (-5)(4),\ (-5)(-11) - (-5)(-4)) = (-20, 20, 35)$, with length $\sqrt{400 + 400 + 1225} =$ 45, so area $BCD = \frac{45}{2}$.

Taking BCD as the base, $10 = \frac{1}{3} \cdot \frac{45}{2} \cdot d$, so $d = \frac{60}{45} = \frac{4}{3}$.

Check: the plane BCD is $4x - 4y - 7z + 4 = 0$ (it contains B, C, D); for A: $\frac{|20 - 12 - 0 + 4|}{\sqrt{16 + 16 + 49}} =$

$$\frac{12}{9} = \frac{4}{3}.$$

Correct Option: (A)

Question 06

English

Let L_1 be the line through the origin parallel to the vector $\hat{i}+\hat{j}+\hat{k}$. For each real number t, let M_t be the line through the point $P_t(t, 2, -1)$ parallel to the vector $\hat{i}+(t-1)\hat{j}-\hat{k}$. The lines L_1 and M_t are coplanar for exactly two values t_1 and t_2 of t. For $i = 1, 2$, let Q_i be the point of intersection of L_1 and M_{t_i}. Then the volume of the tetrahedron with vertices P_{t_1}, P_{t_2}, Q_1 and Q_2 is

(A) $\frac{5}{2}$

(B) $\frac{25}{4}$

(C) $\frac{25}{2}$

(D) $\frac{75}{4}$

Only one option is correct.

हिन्दी

माना L_1 मूलबिन्दु से होकर सदिश $\hat{i}+\hat{j}+\hat{k}$ के समान्तर रेखा है। प्रत्येक वास्तविक संख्या t के लिए माना M_t बिन्दु $P_t(t, 2, -1)$ से होकर सदिश $\hat{i}+(t-1)\hat{j}-\hat{k}$ के समान्तर रेखा है। रेखाएँ L_1 तथा M_t, t के ठीक दो मानों t_1 तथा t_2 के लिए समतलीय हैं। $i = 1, 2$ के लिए माना Q_i रेखाओं L_1 तथा M_{t_i} का प्रतिच्छेदन बिन्दु है। तब शीर्षों P_{t_1}, P_{t_2}, Q_1 तथा Q_2 वाले चतुष्फलक का आयतन है

(A) $\frac{5}{2}$

(B) $\frac{25}{4}$

(C) $\frac{25}{2}$

(D) $\frac{75}{4}$

केवल एक विकल्प सही है।

Solution

Two lines through $\overrightarrow{a}_1$, $\overrightarrow{a}_2$ with directions $\overrightarrow{b}_1$, $\overrightarrow{b}_2$ are coplanar if and only if $\left[\overrightarrow{a}_2 - \overrightarrow{a}_1 \quad \overrightarrow{b}_1 \quad \overrightarrow{b}_2\right] = 0$.

Here $\overrightarrow{a}_1 = \overrightarrow{0}$, $\overrightarrow{b}_1 = (1, 1, 1)$, $\overrightarrow{a}_2 = (t, 2, -1)$, $\overrightarrow{b}_2 = (1, t-1, -1)$.

$$\begin{vmatrix} t & 2 & -1 \\ 1 & 1 & 1 \\ 1 & t-1 & -1 \end{vmatrix} = t\,(-1-(t-1)) - 2(-1-1) - 1\,((t-1)-1) = -t^2 + 4 - t + 2 = -t^2 - t + 6.$$

$-t^2 - t + 6 = 0 \iff (t-2)(t+3) = 0$, so t_1, t_2 are 2 and -3. The direction $(1, t-1, -1)$ is never a multiple of $(1, 1, 1)$ (third component), so in both cases the coplanar lines intersect.

$t = 2$: M_2 is $(2+u,\ 2+u,\ -1-u)$; equating with (s, s, s) gives $2+u = -1-u$, so $u = -\frac{3}{2}$, $s = \frac{1}{2}$ and $Q = \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$.

$t = -3$: M_{-3} is $(-3+u,\ 2-4u,\ -1-u)$; $-3+u = -1-u$ gives $u = 1$, and then $2-4u = -2$ agrees, so $Q = (-2, -2, -2)$.

The volume does not depend on which root is called t_1. Take the vertex $P_2 = (2, 2, -1)$; the other vertices give $\overrightarrow{P_2P_{-3}} = (-5, 0, 0)$, $\overrightarrow{P_2Q} = \left(-\frac{3}{2}, -\frac{3}{2}, \frac{3}{2}\right)$ for $Q = \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$, and $(-4, -4, -1)$ for $Q = (-2, -2, -2)$.

$$\begin{vmatrix} -5 & 0 & 0 \\ -\frac{3}{2} & -\frac{3}{2} & \frac{3}{2} \\ -4 & -4 & -1 \end{vmatrix} = -5\left(\frac{3}{2} + 6\right) = -\frac{75}{2}.$$

Volume $= \frac{1}{6} \cdot \frac{75}{2} = \frac{25}{4}$.

Correct Option: (B)

Question 07

English

Let $\vec{r}$ be a vector such that $\vec{r} \times \left(\hat{i} + 2\hat{j} - \hat{k}\right) = -4\hat{i} + 3\hat{j} + 2\hat{k}$ and $\vec{r} \times \left(2\hat{i} - \hat{j} + \hat{k}\right) = 3\hat{i} - 6\hat{k}$. Then $|\vec{r}|$ is equal to

(A) $\sqrt{5}$
(B) $\sqrt{21}$
(C) $\sqrt{14}$
(D) 3

Only one option is correct.

हिन्दी

माना $\vec{r}$ एक सदिश है जिसके लिए $\vec{r} \times \left(\hat{i} + 2\hat{j} - \hat{k}\right) = -4\hat{i} + 3\hat{j} + 2\hat{k}$ तथा $\vec{r} \times \left(2\hat{i} - \hat{j} + \hat{k}\right) = 3\hat{i} - 6\hat{k}$ है। तब $|\vec{r}|$ बराबर है

(A) $\sqrt{5}$
(B) $\sqrt{21}$
(C) $\sqrt{14}$
(D) 3

केवल एक विकल्प सही है।

Solution

Let $\vec{a} = \hat{i} + 2\hat{j} - \hat{k}$, $\vec{c} = 2\hat{i} - \hat{j} + \hat{k}$ and $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$.

Note first that $\vec{a} \cdot (-4, 3, 2) = -4 + 6 - 2 = 0$ and $\vec{c} \cdot (3, 0, -6) = 6 - 6 = 0$, as needed for each equation to have solutions.

$$\vec{r} \times \vec{a} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x & y & z \\ 1 & 2 & -1 \end{vmatrix} = (-y - 2z)\hat{i} + (x + z)\hat{j} + (2x - y)\hat{k}, \text{ so } -y - 2z = -4,\ x + z = 3,\ 2x - y = 2.$$

$$\vec{r} \times \vec{c} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x & y & z \\ 2 & -1 & 1 \end{vmatrix} = (y + z)\hat{i} + (2z - x)\hat{j} + (-x - 2y)\hat{k}, \text{ so } y + z = 3,\ 2z - x = 0,\ -x - 2y = -6.$$

Each triple alone has rank two: the first describes the line $\vec{r} = \hat{i} + 2\hat{k} + s\,\vec{a}$ (since $(\hat{i} + 2\hat{k}) \times \vec{a} = -4\hat{i} + 3\hat{j} + 2\hat{k}$), the second the line $\vec{r} = -2\hat{i} + 4\hat{j} - \hat{k} + u\,\vec{c}$.

Together: $x = 2z$ and $x + z = 3$ give $z = 1$, $x = 2$; then $2x - y = 2$ gives $y = 2$.

Check all six: $-2-2 = -4$, $2+1 = 3$, $4-2 = 2$, $2+1 = 3$, $2-2 = 0$, $-2-4 = -6$. So $\vec{r} = 2\hat{i}+2\hat{j}+\hat{k}$ is the unique common point of the two lines ($s = 1$, $u = 2$).

$|\overrightarrow{r}| = \sqrt{4+4+1} = 3$. (Options (A) and (B) are the lengths of the base points $\hat{i}+2\hat{k}$ and $-2\hat{i}+4\hat{j}-\hat{k}$ of the two lines.)

Correct Option: (D)

Question 08

English

Let $ABCD$ be a tetrahedron in which $AB = 2$, $AC = 3$, $AD = 4$ and each of the angles $\angle BAC$, $\angle CAD$ and $\angle DAB$ is equal to $60°$. Let $\overrightarrow{u} = \overrightarrow{AB}$, $\overrightarrow{v} = \overrightarrow{AC}$ and $\overrightarrow{w} = \overrightarrow{AD}$. Then the value of

$$(\overrightarrow{u} \times \overrightarrow{v}) \cdot (\overrightarrow{v} \times \overrightarrow{w}) + (\overrightarrow{v} \times \overrightarrow{w}) \cdot (\overrightarrow{w} \times \overrightarrow{u}) + (\overrightarrow{w} \times \overrightarrow{u}) \cdot (\overrightarrow{u} \times \overrightarrow{v})$$

is

(A) 54
(B) -27
(C) -54
(D) -18

Only one option is correct.

हिन्दी

माना $ABCD$ एक चतुष्फलक है जिसमें $AB = 2$, $AC = 3$, $AD = 4$ तथा प्रत्येक कोण $\angle BAC$, $\angle CAD$ तथा $\angle DAB$, $60°$ के बराबर है। माना $\overrightarrow{u} = \overrightarrow{AB}$, $\overrightarrow{v} = \overrightarrow{AC}$ तथा $\overrightarrow{w} = \overrightarrow{AD}$ है। तब

$$(\overrightarrow{u} \times \overrightarrow{v}) \cdot (\overrightarrow{v} \times \overrightarrow{w}) + (\overrightarrow{v} \times \overrightarrow{w}) \cdot (\overrightarrow{w} \times \overrightarrow{u}) + (\overrightarrow{w} \times \overrightarrow{u}) \cdot (\overrightarrow{u} \times \overrightarrow{v})$$

का मान है

(A) 54
(B) -27
(C) -54
(D) -18

केवल एक विकल्प सही है।

Solution

For any four vectors, $(\overrightarrow{a} \times \overrightarrow{b}) \cdot (\overrightarrow{c} \times \overrightarrow{d}) = \begin{vmatrix} \overrightarrow{a} \cdot \overrightarrow{c} & \overrightarrow{a} \cdot \overrightarrow{d} \\ \overrightarrow{b} \cdot \overrightarrow{c} & \overrightarrow{b} \cdot \overrightarrow{d} \end{vmatrix}$, so no coordinates are needed.

Dot products from the data: $\overrightarrow{u} \cdot \overrightarrow{v} = 2 \cdot 3 \cos 60° = 3$, $\overrightarrow{v} \cdot \overrightarrow{w} = 3 \cdot 4 \cos 60° = 6$, $\overrightarrow{w} \cdot \overrightarrow{u} = 4 \cdot 2 \cos 60° = 4$, and $|\overrightarrow{u}|^2 = 4$, $|\overrightarrow{v}|^2 = 9$, $|\overrightarrow{w}|^2 = 16$.

First term ($\overrightarrow{a} = \overrightarrow{u}$, $\overrightarrow{b} = \overrightarrow{v}$, $\overrightarrow{c} = \overrightarrow{v}$, $\overrightarrow{d} = \overrightarrow{w}$): $(\overrightarrow{u} \cdot \overrightarrow{v})(\overrightarrow{v} \cdot \overrightarrow{w}) - (\overrightarrow{u} \cdot \overrightarrow{w})(\overrightarrow{v} \cdot \overrightarrow{v}) = 3 \cdot 6 - 4 \cdot 9 = -18$.

Second term: $(\overrightarrow{v} \cdot \overrightarrow{w})(\overrightarrow{w} \cdot \overrightarrow{u}) - (\overrightarrow{v} \cdot \overrightarrow{u})(\overrightarrow{w} \cdot \overrightarrow{w}) = 6 \cdot 4 - 3 \cdot 16 = -24$.

Third term: $(\overrightarrow{w} \cdot \overrightarrow{u})(\overrightarrow{u} \cdot \overrightarrow{v}) - (\overrightarrow{w} \cdot \overrightarrow{v})(\overrightarrow{u} \cdot \overrightarrow{u}) = 4 \cdot 3 - 6 \cdot 4 = -12$.

Sum $= -18 - 24 - 12 = -54$.

Check with coordinates: $\overrightarrow{u} = (2, 0, 0)$, $\overrightarrow{v} = \left(\frac{3}{2}, \frac{3\sqrt{3}}{2}, 0\right)$, $\overrightarrow{w} = \left(2, \frac{2}{\sqrt{3}}, \frac{4\sqrt{2}}{\sqrt{3}}\right)$ have the same lengths and angles and give the same sum -54.

Correct Option: (C)

Question 09

English

Let P_1 and P_2 be the planes $x + 2y + 2z - 5 = 0$ and $2x + 3y + 6z + 4 = 0$, respectively. For a real number k, let Π_k denote the plane $(x + 2y + 2z - 5) + k(2x + 3y + 6z + 4) = 0$. Let k_0 be the value of k for which the plane Π_k passes through the point $(3, 8, -1)$. Then which of the following statements is TRUE?

(A) $k_0 = -\frac{3}{7}$ and Π_{k_0} bisects the obtuse angle between P_1 and P_2

(B) $k_0 = -\frac{3}{7}$ and Π_{k_0} bisects the acute angle between P_1 and P_2

(C) $k_0 = \frac{3}{7}$ and Π_{k_0} bisects the obtuse angle between P_1 and P_2

(D) $k_0 = -\frac{3}{7}$ and Π_{k_0} does not bisect any angle between P_1 and P_2

Only one option is correct.

हिन्दी

माना P_1 तथा P_2 क्रमशः समतल $x + 2y + 2z - 5 = 0$ तथा $2x + 3y + 6z + 4 = 0$ हैं। एक वास्तविक संख्या k के लिए माना Π_k समतल $(x + 2y + 2z - 5) + k(2x + 3y + 6z + 4) = 0$ को निरूपित करता है। माना k_0, k का वह मान है जिसके लिए समतल Π_k बिन्दु $(3, 8, -1)$ से होकर जाता है। तब निम्न में से कौन-सा कथन सत्य है?

(A) $k_0 = -\frac{3}{7}$ तथा Π_{k_0}, P_1 तथा P_2 के बीच के अधिक कोण (obtuse angle) को समद्विभाजित करता है

(B) $k_0 = -\frac{3}{7}$ तथा Π_{k_0}, P_1 तथा P_2 के बीच के न्यून कोण को समद्विभाजित करता है

(C) $k_0 = \frac{3}{7}$ तथा Π_{k_0}, P_1 तथा P_2 के बीच के अधिक कोण को समद्विभाजित करता है

(D) $k_0 = -\frac{3}{7}$ तथा Π_{k_0}, P_1 तथा P_2 के बीच के किसी भी कोण को समद्विभाजित नहीं करता

केवल एक विकल्प सही है।

Solution

The planes Π_k form the family of planes through the line of intersection of P_1 and P_2.

Substituting $(3, 8, -1)$: $x+2y+2z-5 = 3+16-2-5 = 12$ and $2x+3y+6z+4 = 6+24-6+4 = 28$. So $12 + 28k_0 = 0$ and $k_0 = -\frac{12}{28} = -\frac{3}{7}$.

The normals $\overrightarrow{n}_1 = (1, 2, 2)$, $\overrightarrow{n}_2 = (2, 3, 6)$ have lengths 3 and 7. The bisector planes of P_1, P_2 are $\frac{x + 2y + 2z - 5}{3} = \pm\frac{2x + 3y + 6z + 4}{7}$, i.e. the members $k = \mp\frac{3}{7}$ of the family.

So Π_{k_0}, which is $\frac{x + 2y + 2z - 5}{3} = \frac{2x + 3y + 6z + 4}{7}$, is one of the two bisector planes; option (D) is wrong, and option (C) has the wrong sign of k_0.

Which angle? With unit normals $\hat{n}_1 = \frac{\overrightarrow{n}_1}{3}$, $\hat{n}_2 = \frac{\overrightarrow{n}_2}{7}$: $\hat{n}_1 \cdot \hat{n}_2 = \frac{2 + 6 + 12}{21} = \frac{20}{21}$, so the acute angle θ between P_1, P_2 satisfies $\cos\theta = \frac{20}{21}$.

Π_{k_0} is $(\hat{n}_1 - \hat{n}_2) \cdot \overrightarrow{r} = \text{const}$; its normal is $\hat{n}_1 - \hat{n}_2$, with $|\hat{n}_1 - \hat{n}_2|^2 = 2 - \frac{40}{21} = \frac{2}{21}$ and $\hat{n}_1 \cdot (\hat{n}_1 - \hat{n}_2) = 1 - \frac{20}{21} = \frac{1}{21}$.

The angle ϕ between Π_{k_0} and P_1: $\cos\phi = \frac{1/21}{\sqrt{2/21}} = \frac{1}{\sqrt{42}}$, so $\cos 2\phi = 2\cos^2\phi - 1 = \frac{1}{21} - 1 = -\frac{20}{21} = \cos(\pi - \theta)$.

Hence Π_{k_0} makes the angle $\dfrac{\pi-\theta}{2}$ with each plane: it bisects the obtuse angle. (Equivalently, since $\overrightarrow{n}_1 \cdot \overrightarrow{n}_2 > 0$, the member with normal $\hat{n}_1 + \hat{n}_2$, i.e. $k = \dfrac{3}{7}$, bisects the acute angle, so (B) is wrong.)

Correct Option: (A)

Question 10

English

Let $P_1 : 2x + y + 2z = 4$ and $P_2 : x + 2y - 2z = -2$ be two planes, and let Π be the plane which bisects that angle between P_1 and P_2 which contains the origin. Let L_1 and L_2 be the lines in which P_1 and P_2, respectively, meet the plane $z = 0$. In the plane $z = 0$ there are exactly two circles which touch both L_1 and L_2 and have their centres on the x-axis; let C be the one whose centre lies on Π. Then the radius of C is

(A) $\dfrac{8\sqrt{5}}{5}$

(B) $\dfrac{8}{9}$

(C) $\dfrac{8\sqrt{5}}{15}$

(D) $\dfrac{16\sqrt{5}}{15}$

Only one option is correct.

हिन्दी

माना $P_1 : 2x + y + 2z = 4$ तथा $P_2 : x + 2y - 2z = -2$ दो समतल हैं, तथा Π वह समतल है जो P_1 तथा P_2 के बीच के उस कोण को समद्विभाजित करता है जिसमें मूलबिन्दु स्थित है। माना L_1 तथा L_2 वे रेखाएँ हैं जिनमें क्रमशः P_1 तथा P_2 समतल $z = 0$ से मिलते हैं। समतल $z = 0$ में ठीक दो वृत्त ऐसे हैं जो L_1 तथा L_2 दोनों को स्पर्श करते हैं तथा जिनके केन्द्र x-अक्ष पर हैं; माना C उनमें से वह वृत्त है जिसका केन्द्र Π पर स्थित है। तब C की त्रिज्या है

(A) $\dfrac{8\sqrt{5}}{5}$

(B) $\dfrac{8}{9}$

(C) $\dfrac{8\sqrt{5}}{15}$

(D) $\dfrac{16\sqrt{5}}{15}$

केवल एक विकल्प सही है।

Solution

Write $f_1 = 2x + y + 2z - 4$ and $f_2 = x + 2y - 2z + 2$, so $P_1 : f_1 = 0$, $P_2 : f_2 = 0$. Both normals $(2, 1, 2)$ and $(1, 2, -2)$ have length 3.

At the origin $f_1 = -4 < 0$ and $f_2 = 2 > 0$. The angle between the planes that contains O is the region where f_1 and f_2 keep these signs, and its bisector is the set of points there with equal distances: $\dfrac{-f_1}{3} = \dfrac{f_2}{3}$.

So $\Pi : f_1 + f_2 = 0$, i.e. $3x + 3y - 2 = 0$. (The other bisector is $f_1 = f_2$, i.e. $x - y + 4z - 6 = 0$.)

Traces in the plane $z = 0$: $L_1 : 2x + y - 4 = 0$ and $L_2 : x + 2y + 2 = 0$.

A circle in $z = 0$ touching L_1 and L_2 has its centre equidistant from them, hence on one of the two angle bisectors of L_1 and L_2. For a centre $(h, 0)$ on the x-axis: $\dfrac{|2h - 4|}{\sqrt{5}} = \dfrac{|h + 2|}{\sqrt{5}}$.

$2h - 4 = h + 2 \Rightarrow h = 6$, or $2h - 4 = -(h + 2) \Rightarrow h = \dfrac{2}{3}$. These are the two circles of the question.

The centre $(h, 0, 0)$ lies on Π only for $3h - 2 = 0$, i.e. $h = \dfrac{2}{3}$. (Indeed the trace of Π on $z = 0$ is $x + y = \dfrac{2}{3}$, which is the bisector $\dfrac{2x + y - 4}{\sqrt{5}} = -\dfrac{x + 2y + 2}{\sqrt{5}}$ of L_1, L_2 through the region of O, because both normalising factors are equal.)

Radius = distance from $\left(\dfrac{2}{3}, 0\right)$ to $L_1 = \dfrac{\left|\frac{4}{3} - 4\right|}{\sqrt{5}} = \dfrac{8}{3\sqrt{5}} = \dfrac{8\sqrt{5}}{15}$, and the distance to L_2 is $\dfrac{\left|\frac{2}{3} + 2\right|}{\sqrt{5}}$, the same.

(Option (A) is the radius of the other circle, centre $(6, 0)$; option (B) uses the 3D normalising factor 3 instead of $\sqrt{5}$.)

Correct Option: (C)

Question 11

English

Let S_0 be the sphere $x^2 + y^2 + z^2 = 18$ and let C be the circle in which S_0 meets the plane $x + 2y + 2z = 9$. Let L be the line $x = -1,\ y + 3 = z - 2$. There are exactly two spheres each of which contains the circle C and touches the line L. Then the radius of the smaller of these two spheres is

(A) 3
(B) $\dfrac{15}{4}$
(C) $\dfrac{3\sqrt{41}}{4}$
(D) $3\sqrt{2}$

Only one option is correct.

हिन्दी

माना S_0 गोला $x^2 + y^2 + z^2 = 18$ है तथा C वह वृत्त है जिसमें S_0 समतल $x + 2y + 2z = 9$ से मिलता है। माना L रेखा $x = -1,\ y + 3 = z - 2$ है। ठीक दो गोले ऐसे हैं जिनमें से प्रत्येक वृत्त C को समाहित करता है तथा रेखा L को स्पर्श करता है। तब इन दोनों गोलों में से छोटे गोले की त्रिज्या है

(A) 3
(B) $\dfrac{15}{4}$
(C) $\dfrac{3\sqrt{41}}{4}$
(D) $3\sqrt{2}$

केवल एक विकल्प सही है।

Solution

The distance of the centre O of S_0 from the plane $x + 2y + 2z = 9$ is $\dfrac{9}{\sqrt{1 + 4 + 4}} = 3 < 3\sqrt{2}$, so the plane cuts S_0 in a circle.

The centre of C is the foot N of the perpendicular from O to the plane: $N = \dfrac{9}{9}(1, 2, 2) = (1, 2, 2)$; its radius is $\sqrt{18 - 9} = 3$.

Every point of C is at distance 3 from N in the plane, so a point K is equidistant from all points of C exactly when K lies on the line through N perpendicular to the plane. So the centre of a sphere

containing C is $K = (1+\lambda,\ 2+2\lambda,\ 2+2\lambda)$, and its radius satisfies $R^2 = NK^2 + 3^2 = 9\lambda^2 + 9$.
The line L passes through $A(-1,-3,2)$ with direction $\overrightarrow{e} = (0,1,1)$, $|\overrightarrow{e}| = \sqrt{2}$.
$\overrightarrow{AK} = (2+\lambda,\ 5+2\lambda,\ 2\lambda)$ and $\overrightarrow{AK} \times \overrightarrow{e} = ((5+2\lambda) - 2\lambda,\ -(2+\lambda),\ 2+\lambda) = (5,\ -2-\lambda,\ 2+\lambda)$.
Distance of K from L: $d^2 = \dfrac{25 + 2(2+\lambda)^2}{2}$.
The sphere touches L when $d = R$: $25 + 2(4 + 4\lambda + \lambda^2) = 18 + 18\lambda^2$, i.e. $16\lambda^2 - 8\lambda - 15 = 0$, so $\lambda = \dfrac{8 \pm 32}{32}$: $\lambda = \dfrac{5}{4}$ or $\lambda = -\dfrac{3}{4}$.
$\lambda = -\dfrac{3}{4}$: $R^2 = 9 + 9 \cdot \dfrac{9}{16} = \dfrac{225}{16}$, $R = \dfrac{15}{4}$, centre $\left(\dfrac{1}{4}, \dfrac{1}{2}, \dfrac{1}{2}\right)$.
$\lambda = \dfrac{5}{4}$: $R^2 = 9 + 9 \cdot \dfrac{25}{16} = \dfrac{369}{16}$, $R = \dfrac{3\sqrt{41}}{4}$. ($S_0$ itself, $\lambda = -1$, does not touch L: its centre is at distance $\sqrt{\frac{27}{2}}$ from L, not $3\sqrt{2}$.)
The smaller radius is $\dfrac{15}{4}$.

Correct Option: (B)

Question 12

English

Let S be the sphere $x^2 + y^2 + z^2 - 2x + 4y - 6z - 11 = 0$ and let Π be the plane $x + y - z = 1$. The plane Π passes through the point $P(5,1,5)$ and meets S in a circle C. Then the length of a tangent drawn from P to the circle C, in the plane Π, is

(A) $\sqrt{29}$
(B) $\dfrac{5\sqrt{6}}{3}$
(C) $2\sqrt{2}$
(D) 2

Only one option is correct.

हिन्दी

माना S गोला $x^2 + y^2 + z^2 - 2x + 4y - 6z - 11 = 0$ है तथा Π समतल $x + y - z = 1$ है। समतल Π बिन्दु $P(5,1,5)$ से होकर जाता है तथा S से एक वृत्त C में मिलता है। तब समतल Π में P से वृत्त C पर खींची गई स्पर्श रेखा की लम्बाई है

(A) $\sqrt{29}$
(B) $\dfrac{5\sqrt{6}}{3}$
(C) $2\sqrt{2}$
(D) 2

केवल एक विकल्प सही है।

Solution

S: $(x-1)^2 + (y+2)^2 + (z-3)^2 = 1 + 4 + 9 + 11 = 25$, centre $K(1,-2,3)$, radius 5. The point $P(5,1,5)$ satisfies $5 + 1 - 5 = 1$, so it lies on Π.
Distance of K from Π: $h = \dfrac{|1 + (-2) - 3 - 1|}{\sqrt{1+1+1}} = \dfrac{5}{\sqrt{3}} < 5$, so Π cuts S in a circle.

The centre N of C is the foot of the perpendicular from K to Π: $N = K - \dfrac{(1-2-3)-1}{3}(1,1,-1) = K + \dfrac{5}{3}(1,1,-1) = \left(\dfrac{8}{3}, -\dfrac{1}{3}, \dfrac{4}{3}\right)$.

Radius of C: $\rho^2 = 25 - h^2 = 25 - \dfrac{25}{3} = \dfrac{50}{3}$.

$\overrightarrow{NP} = \left(5 - \dfrac{8}{3},\ 1 + \dfrac{1}{3},\ 5 - \dfrac{4}{3}\right) = \left(\dfrac{7}{3}, \dfrac{4}{3}, \dfrac{11}{3}\right)$, so $PN^2 = \dfrac{49 + 16 + 121}{9} = \dfrac{62}{3} > \rho^2$: P is outside C.

In the plane Π, the tangent from P to C has length t with $t^2 = PN^2 - \rho^2 = \dfrac{62}{3} - \dfrac{50}{3} = 4$, so $t = 2$.

Check by the power of P with respect to S: $KN \perp \Pi$ gives $PK^2 = PN^2 + h^2$, hence $PN^2 - \rho^2 = PK^2 - h^2 - (25 - h^2) = PK^2 - 25 = 16 + 9 + 4 - 25 = 4$. The tangent length is the same for every plane through P that cuts S.

Correct Option: (D)

Question 13

English

Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ be a vector of magnitude 7 whose components x, y, z are non-zero integers. Let α, β, γ be the angles which $\vec{r}$ makes with the positive directions of the x-axis, the y-axis and the z-axis respectively. It is given that α and γ are obtuse, $\beta < \dfrac{\pi}{3}$ and $\alpha > \gamma$. Then the value of x is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ परिमाण 7 का एक सदिश है जिसके घटक x, y, z शून्येतर पूर्णांक हैं। माना α, β, γ वे कोण हैं जो $\vec{r}$ क्रमशः x-अक्ष, y-अक्ष तथा z-अक्ष की धनात्मक दिशाओं के साथ बनाता है। दिया है कि α तथा γ अधिक कोण हैं, $\beta < \dfrac{\pi}{3}$ है तथा $\alpha > \gamma$ है। तब x का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Since $|\vec{r}| = 7$, we need $x^2 + y^2 + z^2 = 49$ with x, y, z non-zero integers, so each square is one of $1, 4, 9, 16, 25, 36$.

If the largest square is 36, the other two add to $13 = 4 + 9$ (no other split of 13 into two non-zero squares).

If the largest square is 25, the other two add to 24; but $24 - 1 = 23$, $24 - 4 = 20$, $24 - 9 = 15$, $24 - 16 = 8$ are not squares, so this is impossible.

If the largest square is at most 16, the sum is at most $48 < 49$. Hence $\{|x|, |y|, |z|\} = \{2, 3, 6\}$.

The direction cosines are $\cos\alpha = \dfrac{x}{7}$, $\cos\beta = \dfrac{y}{7}$, $\cos\gamma = \dfrac{z}{7}$.

α and γ obtuse give $\cos\alpha < 0$ and $\cos\gamma < 0$, so $x < 0$ and $z < 0$.

$\beta < \dfrac{\pi}{3}$ gives $\cos\beta > \cos\dfrac{\pi}{3} = \dfrac{1}{2}$, so $y > 3.5$; from $\{2, 3, 6\}$ this forces $y = 6$, and then $\{|x|, |z|\} = \{2, 3\}$.

The angles lie in $[0, \pi]$, where cosine is decreasing; $\alpha > \gamma$ therefore means $\cos\alpha < \cos\gamma$, that is $x < z$.

With $x, z \in \{-2, -3\}$ and $x < z$: $x = -3$, $z = -2$, so $\vec{r} = -3\hat{i} + 6\hat{j} - 2\hat{k}$ (check: $9 + 36 + 4 = 49$).

Final Answer: -3

Question 14

English

Consider the vectors $\vec{a} = \hat{i} + 2\hat{j} + 2\hat{k}$, $\vec{b} = 2\hat{i} - 2\hat{j} + \hat{k}$ and $\vec{c} = -2\hat{i} - \hat{j} + 2\hat{k}$. A particle starts from the origin O and makes five successive straight-line moves. For $k = 1, 2, 3, 4, 5$, in the k-th move it travels a distance of k units in the direction of $\vec{a}$, $\vec{b}$, $\vec{c}$, $\vec{a}$, $\vec{b}$ respectively. If P is the final position of the particle, then the value of OP^2 is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

सदिशों $\vec{a} = \hat{i} + 2\hat{j} + 2\hat{k}$, $\vec{b} = 2\hat{i} - 2\hat{j} + \hat{k}$ तथा $\vec{c} = -2\hat{i} - \hat{j} + 2\hat{k}$ पर विचार कीजिए। एक कण मूलबिन्दु O से चलना प्रारम्भ करता है तथा क्रमागत पाँच सरल रेखीय चालें चलता है। $k = 1, 2, 3, 4, 5$ के लिए, k-वीं चाल में वह क्रमशः $\vec{a}$, $\vec{b}$, $\vec{c}$, $\vec{a}$, $\vec{b}$ की दिशा में k इकाई की दूरी तय करता है। यदि P कण की अन्तिम स्थिति है, तो OP^2 का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Each given vector has magnitude 3: $|\vec{a}| = \sqrt{1+4+4} = 3$, $|\vec{b}| = \sqrt{4+4+1} = 3$, $|\vec{c}| = \sqrt{4+1+4} = 3$.

A move of k units in the direction of $\vec{v}$ is the displacement $k\hat{v} = \dfrac{k}{|\vec{v}|}\vec{v} = \dfrac{k}{3}\vec{v}$.

The five displacements are $\dfrac{1}{3}\vec{a}$, $\dfrac{2}{3}\vec{b}$, $\dfrac{3}{3}\vec{c}$, $\dfrac{4}{3}\vec{a}$, $\dfrac{5}{3}\vec{b}$.

Adding (vector addition is commutative and associative): $\overrightarrow{OP} = \dfrac{5\vec{a} + 7\vec{b} + 3\vec{c}}{3}$.

$5\vec{a} = 5\hat{i} + 10\hat{j} + 10\hat{k}$, $7\vec{b} = 14\hat{i} - 14\hat{j} + 7\hat{k}$, $3\vec{c} = -6\hat{i} - 3\hat{j} + 6\hat{k}$.

Sum $= 13\hat{i} - 7\hat{j} + 23\hat{k}$, so $\overrightarrow{OP} = \dfrac{13}{3}\hat{i} - \dfrac{7}{3}\hat{j} + \dfrac{23}{3}\hat{k}$.

$OP^2 = \dfrac{13^2 + 7^2 + 23^2}{9} = \dfrac{169 + 49 + 529}{9} = \dfrac{747}{9} = 83.$

(Check: $\vec{a}$, $\vec{b}$, $\vec{c}$ are mutually perpendicular, so $OP^2 = 5^2 + 7^2 + 3^2 = 83$ as well.)

Final Answer: 83

Question 15

English

Let $\hat{u} = l\hat{i} + m\hat{j} + n\hat{k}$ be a unit vector which makes an obtuse angle with the positive direction of the z-axis. Let $w = l + \mathrm{i}\, m$, where $\mathrm{i} = \sqrt{-1}$, be such that $|w| = \dfrac{3}{5}$ and $(3 - 4\mathrm{i})\,\overline{w}$ is a negative real number, where $\overline{w}$ denotes the complex conjugate of w. Then the value of $l + m + n$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $\hat{u} = l\hat{i} + m\hat{j} + n\hat{k}$ एक मात्रक सदिश है जो z-अक्ष की धनात्मक दिशा के साथ अधिक कोण बनाता है। माना $w = l + \mathrm{i}\, m$, जहाँ $\mathrm{i} = \sqrt{-1}$ है, इस प्रकार है कि $|w| = \dfrac{3}{5}$ है तथा $(3 - 4\mathrm{i})\,\overline{w}$ एक ऋणात्मक वास्तविक संख्या है, जहाँ $\overline{w}$, w का संयुग्मी सम्मिश्र है। तब $l + m + n$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Write $w = |w|(\cos\theta + \mathrm{i}\sin\theta)$ with $\theta = \arg w$ and $|w| = \dfrac{3}{5}$; then $\overline{w} = \dfrac{3}{5}(\cos\theta - \mathrm{i}\sin\theta) = \dfrac{3}{5}e^{-\mathrm{i}\theta}$.

$|3 - 4\mathrm{i}| = 5$ and $3 - 4\mathrm{i}$ lies in the fourth quadrant, so $3 - 4\mathrm{i} = 5e^{-\mathrm{i}\varphi}$ with $\varphi = \tan^{-1}\dfrac{4}{3} \in \left(0, \dfrac{\pi}{2}\right)$, $\cos\varphi = \dfrac{3}{5}$, $\sin\varphi = \dfrac{4}{5}$.

Then $(3 - 4\mathrm{i})\overline{w} = 3e^{-\mathrm{i}(\theta+\varphi)}$. This is a negative real number exactly when $\theta + \varphi$ is an odd multiple of π, so $\theta = \pi - \varphi$ up to multiples of 2π.

Hence $\cos\theta = -\cos\varphi = -\dfrac{3}{5}$ and $\sin\theta = \sin\varphi = \dfrac{4}{5}$.

So $w = \dfrac{3}{5}\left(-\dfrac{3}{5} + \dfrac{4}{5}\mathrm{i}\right) = -\dfrac{9}{25} + \dfrac{12}{25}\mathrm{i}$, giving $l = -0.36$ and $m = 0.48$.

Check: $(3 - 4\mathrm{i})\left(-\dfrac{9}{25} - \dfrac{12}{25}\mathrm{i}\right) = \dfrac{-27 - 36\mathrm{i} + 36\mathrm{i} + 48\mathrm{i}^2}{25} = -3 < 0$.

Since $\hat{u}$ is a unit vector, $l^2 + m^2 + n^2 = 1$, and $l^2 + m^2 = |w|^2 = \dfrac{9}{25}$, so $n^2 = \dfrac{16}{25}$.

The angle with the positive z-axis is obtuse, so $n = \cos\gamma < 0$, that is $n = -\dfrac{4}{5} = -0.8$.

Therefore $l + m + n = -0.36 + 0.48 - 0.8 = -0.68$.

Final Answer: -0.68

Question 16

English

Let $\vec{a}_0,\ \vec{b}_0,\ \vec{c}_0$ be three non-coplanar vectors. For every integer $n \geq 0$, define $\vec{a}_{n+1} = \dfrac{\vec{b}_n + \vec{c}_n}{2}$, $\vec{b}_{n+1} = \dfrac{\vec{c}_n + \vec{a}_n}{2}$ and $\vec{c}_{n+1} = \dfrac{\vec{a}_n + \vec{b}_n}{2}$. If $\vec{c}_3 = \lambda\vec{a}_0 + \mu\vec{b}_0 + \nu\vec{c}_0$, where λ, μ, ν are real numbers, then the value of ν is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $\vec{a}_0,\ \vec{b}_0,\ \vec{c}_0$ तीन असमतलीय सदिश हैं। प्रत्येक पूर्णांक $n \geq 0$ के लिए $\vec{a}_{n+1} = \dfrac{\vec{b}_n + \vec{c}_n}{2}$, $\vec{b}_{n+1} = \dfrac{\vec{c}_n + \vec{a}_n}{2}$ तथा $\vec{c}_{n+1} = \dfrac{\vec{a}_n + \vec{b}_n}{2}$ परिभाषित कीजिए। यदि $\vec{c}_3 = \lambda\vec{a}_0 + \mu\vec{b}_0 + \nu\vec{c}_0$ है, जहाँ λ, μ, ν वास्तविक संख्याएँ हैं, तो ν का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Adding the three defining relations gives $\vec{a}_{n+1} + \vec{b}_{n+1} + \vec{c}_{n+1} = \vec{a}_n + \vec{b}_n + \vec{c}_n$, so $\vec{S} = \vec{a}_0 + \vec{b}_0 + \vec{c}_0$ equals $\vec{a}_n + \vec{b}_n + \vec{c}_n$ for every n.

Hence $\vec{c}_{n+1} = \dfrac{\vec{a}_n + \vec{b}_n}{2} = \dfrac{\vec{S} - \vec{c}_n}{2}$.

Subtracting $\dfrac{\vec{S}}{3}$ from both sides: $\vec{c}_{n+1} - \dfrac{\vec{S}}{3} = \dfrac{\vec{S}}{6} - \dfrac{\vec{c}_n}{2} = -\dfrac{1}{2}\left(\vec{c}_n - \dfrac{\vec{S}}{3}\right)$.

Applying this three times: $\vec{c}_3 - \dfrac{\vec{S}}{3} = \left(-\dfrac{1}{2}\right)^3\left(\vec{c}_0 - \dfrac{\vec{S}}{3}\right) = -\dfrac{1}{8}\vec{c}_0 + \dfrac{1}{24}\vec{S}$.

So $\vec{c}_3 = \dfrac{\vec{S}}{3} + \dfrac{\vec{S}}{24} - \dfrac{\vec{c}_0}{8} = \dfrac{3}{8}\vec{S} - \dfrac{1}{8}\vec{c}_0 = \dfrac{3}{8}\vec{a}_0 + \dfrac{3}{8}\vec{b}_0 + \dfrac{1}{4}\vec{c}_0$.

Direct check: $\vec{a}_2 = \dfrac{\vec{b}_1 + \vec{c}_1}{2} = \dfrac{2\vec{a}_0 + \vec{b}_0 + \vec{c}_0}{4}$, $\vec{b}_2 = \dfrac{\vec{a}_0 + 2\vec{b}_0 + \vec{c}_0}{4}$, and $\vec{c}_3 = \dfrac{\vec{a}_2 + \vec{b}_2}{2} = \dfrac{3\vec{a}_0 + 3\vec{b}_0 + 2\vec{c}_0}{8}$.

Because $\vec{a}_0,\ \vec{b}_0, \vec{c}_0$ are non-coplanar, every vector has a unique expression as their linear combination, so $\lambda = \mu = \dfrac{3}{8}$ and $\nu = \dfrac{1}{4}$.

Final Answer: 0.25

Question 17

English

Let $\vec{a} = 2\hat{i}+4\hat{j}+4\hat{k}$, $\vec{b} = 4\hat{i}+2\hat{j}-4\hat{k}$ and $\vec{c} = 5\hat{i}+2\hat{j}$. For $\theta \in \left[0, \frac{\pi}{2}\right]$, let $\vec{r}(\theta) = \vec{c} + (\cos\theta)\,\vec{a} + (\sin\theta)\,\vec{b}$. Let M and m denote the maximum value and the minimum value, respectively, of $|\vec{r}(\theta)|^2$ for $\theta \in \left[0, \frac{\pi}{2}\right]$. Then the value of $M + m$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $\vec{a} = 2\hat{i} + 4\hat{j} + 4\hat{k}$, $\vec{b} = 4\hat{i} + 2\hat{j} - 4\hat{k}$ तथा $\vec{c} = 5\hat{i} + 2\hat{j}$ हैं। $\theta \in \left[0, \frac{\pi}{2}\right]$ के लिए माना $\vec{r}(\theta) = \vec{c} + (\cos\theta)\,\vec{a} + (\sin\theta)\,\vec{b}$ है। माना $\theta \in \left[0, \frac{\pi}{2}\right]$ के लिए $|\vec{r}(\theta)|^2$ के अधिकतम तथा न्यूनतम मान क्रमशः M तथा m हैं। तब $M + m$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Collecting components, $\vec{r}(\theta) = x\hat{i} + y\hat{j} + z\hat{k}$ with $x = 5 + 2\cos\theta + 4\sin\theta$, $y = 2 + 4\cos\theta + 2\sin\theta$, $z = 4\cos\theta - 4\sin\theta$.

$x^2 = 25 + 4\cos^2\theta + 16\sin^2\theta + 20\cos\theta + 40\sin\theta + 16\sin\theta\cos\theta$.

$y^2 = 4 + 16\cos^2\theta + 4\sin^2\theta + 16\cos\theta + 8\sin\theta + 16\sin\theta\cos\theta$.

$z^2 = 16\cos^2\theta + 16\sin^2\theta - 32\sin\theta\cos\theta$.

Adding: constants 29; $\cos^2\theta$ and $\sin^2\theta$ both have coefficient 36, giving 36; the $\sin\theta\cos\theta$ terms cancel; linear terms $36\cos\theta + 48\sin\theta$.

So $f(\theta) = |\vec{r}(\theta)|^2 = 65 + 36\cos\theta + 48\sin\theta$.

Since $\sqrt{36^2 + 48^2} = 60$, $36\cos\theta + 48\sin\theta = 60\cos(\theta - \varphi)$ with $\cos\varphi = \frac{3}{5}$, $\sin\varphi = \frac{4}{5}$, so $\varphi = \tan^{-1}\frac{4}{3} \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$.

The overall range of $a\cos\theta + b\sin\theta$ is $\left[-\sqrt{a^2+b^2}, \sqrt{a^2+b^2}\right]$, but here θ is restricted, so the attained values must be checked.

Maximum: $\theta = \varphi \in \left[0, \frac{\pi}{2}\right]$ is allowed, so $M = 65 + 60 = 125$.

Minimum: on $\left[0, \frac{\pi}{2}\right]$, $\theta - \varphi$ runs over $\left[-\varphi, \frac{\pi}{2} - \varphi\right]$, which contains 0 and has no point where cosine reaches -1; $\cos(\theta - \varphi)$ decreases as $|\theta - \varphi|$ grows, so the minimum is at the end farther from φ.

$f(0) = 65 + 36 = 101$ and $f\left(\frac{\pi}{2}\right) = 65 + 48 = 113$, so $m = 101$.

Therefore $M + m = 125 + 101 = 226$.

Final Answer: 226

Question 18

English

Let L_1 and L_2 be the lines $\frac{x-2}{1} = \frac{y+1}{2} = \frac{z-3}{2}$ and $\frac{x+1}{2} = \frac{y-4}{-1} = \frac{z}{2}$ respectively. Let P be a point on L_1 and Q be a point on L_2 such that the mid-point of the line segment PQ is $\left(4, \frac{9}{4}, 6\right)$. Then the length of PQ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना L_1 तथा L_2 क्रमशः रेखाएँ $\frac{x-2}{1} = \frac{y+1}{2} = \frac{z-3}{2}$ तथा $\frac{x+1}{2} = \frac{y-4}{-1} = \frac{z}{2}$ हैं। माना L_1 पर एक बिन्दु P तथा L_2 पर एक बिन्दु Q इस प्रकार हैं कि रेखाखण्ड PQ का मध्यबिन्दु $\left(4, \frac{9}{4}, 6\right)$ है। तब PQ की लम्बाई _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Writing each symmetric equation equal to a parameter, a general point of L_1 is $P = (2 + s,\ -1 + 2s,\ 3 + 2s)$ and a general point of L_2 is $Q = (-1 + 2t,\ 4 - t,\ 2t)$, $s, t \in \mathbb{R}$.

By the section (mid-point) formula, the mid-point of PQ is $\left(\frac{1+s+2t}{2}, \frac{3+2s-t}{2}, \frac{3+2s+2t}{2}\right)$.

Equating with $\left(4, \frac{9}{4}, 6\right)$: $s + 2t = 7$, $2s - t = \frac{3}{2}$, $2s + 2t = 9$.

Subtracting the first from the third gives $s = 2$; then $t = \frac{5}{2}$.

These satisfy the second equation: $2(2) - \frac{5}{2} = \frac{3}{2}$. So the three equations are consistent and the solution is unique (the directions of L_1, L_2 are not parallel).

$P = (4, 3, 7)$ and $Q = \left(-1 + 5,\ 4 - \frac{5}{2},\ 5\right) = \left(4, \frac{3}{2}, 5\right)$.

By the distance formula, $PQ = \sqrt{(4-4)^2 + \left(3 - \frac{3}{2}\right)^2 + (7-5)^2} = \sqrt{\frac{9}{4} + 4} = \sqrt{\frac{25}{4}} = \frac{5}{2}$.

Final Answer: 2.5

Question 19

English

Let $A(1,-1,2)$ and $B(2,1,-1)$ be two points and let M be the mid-point of the line segment AB. The xy-plane meets the line AB at the point P. Let Q be the harmonic conjugate of P with respect to A and B. Then the value of $MP \cdot MQ$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $A(1,-1,2)$ तथा $B(2,1,-1)$ दो बिन्दु हैं तथा M रेखाखण्ड AB का मध्यबिन्दु है। xy-तल रेखा AB को बिन्दु P पर मिलता है। माना Q, A तथा B के सापेक्ष P का हरात्मक संयुग्मी है। तब $MP \cdot MQ$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

A point of line AB dividing AB in the ratio $k:1$ has z-coordinate $\dfrac{k(-1)+1(2)}{k+1}$; it lies on the xy-plane when $2-k=0$, that is $k=2$.

Since $k=2>0$, the division is internal: $AP:PB=2:1$, and $P=\dfrac{2B+A}{3}=\left(\dfrac{4+1}{3},\dfrac{2-1}{3},\dfrac{-2+2}{3}\right)=\left(\dfrac{5}{3},\dfrac{1}{3},0\right)$.

The harmonic conjugate Q of P with respect to A, B divides AB externally in the same ratio $2:1$: $Q=\dfrac{2B-A}{2-1}=(4-1,\,2+1,\,-2-2)=(3,3,-4)$.

Check: $QA=\sqrt{4+16+36}=2\sqrt{14}$ and $QB=\sqrt{1+4+9}=\sqrt{14}$, so $QA:QB=2:1$.

The mid-point is $M=\left(\dfrac{3}{2},0,\dfrac{1}{2}\right)$.

$\overrightarrow{MP}=\left(\dfrac{5}{3}-\dfrac{3}{2},\dfrac{1}{3},-\dfrac{1}{2}\right)=\left(\dfrac{1}{6},\dfrac{1}{3},-\dfrac{1}{2}\right)$, so $MP=\sqrt{\dfrac{1+4+9}{36}}=\dfrac{\sqrt{14}}{6}$.

$\overrightarrow{MQ}=\left(\dfrac{3}{2},3,-\dfrac{9}{2}\right)=\dfrac{3}{2}(1,2,-3)$, so $MQ=\dfrac{3}{2}\sqrt{14}$.

$MP\cdot MQ=\dfrac{\sqrt{14}}{6}\cdot\dfrac{3\sqrt{14}}{2}=\dfrac{42}{12}=\dfrac{7}{2}$.

This agrees with the property of a harmonic pair, $MP\cdot MQ=MA^2$, where $MA^2=\dfrac{AB^2}{4}=\dfrac{1+4+9}{4}=\dfrac{7}{2}$.

Final Answer: 3.5

Question 20

English

Let A, B and C be the points with position vectors $\hat{i}+2\hat{j}+2\hat{k}$, $2\hat{i}+\hat{j}-2\hat{k}$ and $\hat{i}-\frac{2}{5}\hat{j}-\frac{14}{5}\hat{k}$ respectively. Then the z-coordinate of the orthocentre of the triangle ABC is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना A, B तथा C वे बिन्दु हैं जिनके स्थिति सदिश क्रमशः $\hat{i}+2\hat{j}+2\hat{k}$, $2\hat{i}+\hat{j}-2\hat{k}$ तथा $\hat{i}-\frac{2}{5}\hat{j}-\frac{14}{5}\hat{k}$ हैं। तब त्रिभुज ABC के लम्बकेन्द्र का z-निर्देशांक _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $\vec{a}$, $\vec{b}$, $\vec{c}$ be the position vectors of A, B, C with respect to the origin O.

$|\vec{a}|^2 = 1+4+4 = 9$, $|\vec{b}|^2 = 4+1+4 = 9$, $|\vec{c}|^2 = 1+\frac{4}{25}+\frac{196}{25} = \frac{225}{25} = 9$. So O is at distance 3 from each vertex.

Equal distances alone do not make O the circumcentre in space; O must also lie in the plane of the triangle. The scalar triple product is

$$\left[\vec{a}\ \vec{b}\ \vec{c}\right] = 1\left(-\frac{14}{5}-\frac{4}{5}\right)-2\left(-\frac{28}{5}+2\right)+2\left(-\frac{4}{5}-1\right) = -\frac{18}{5}+\frac{36}{5}-\frac{18}{5} = 0$$

so O, A, B, C are coplanar and O is the circumcentre of triangle ABC.

The centroid has position vector $\vec{g} = \dfrac{\vec{a}+\vec{b}+\vec{c}}{3}$.

On the Euler line, the centroid divides the segment from the circumcentre O to the orthocentre H internally in the ratio $1:2$, so $\vec{g} = \dfrac{2\cdot\vec{0}+1\cdot\vec{h}}{3}$, giving $\vec{h} = \vec{a}+\vec{b}+\vec{c}$.

$\vec{h} = (1+2+1)\hat{i}+\left(2+1-\frac{2}{5}\right)\hat{j}+\left(2-2-\frac{14}{5}\right)\hat{k} = 4\hat{i}+\frac{13}{5}\hat{j}-\frac{14}{5}\hat{k}$.

Check: $\vec{h}-\vec{a} = 3\hat{i}+\frac{3}{5}\hat{j}-\frac{24}{5}\hat{k}$ and $\vec{b}-\vec{c} = \hat{i}+\frac{7}{5}\hat{j}+\frac{4}{5}\hat{k}$; their dot product is $3+\frac{21}{25}-\frac{96}{25} = 0$, so $AH \perp BC$. Similarly $(\vec{h}-\vec{b})\cdot(\vec{c}-\vec{a}) = 2\cdot 0+\frac{8}{5}\left(-\frac{12}{5}\right)+\left(-\frac{4}{5}\right)\left(-\frac{24}{5}\right) = 0$.

The z-coordinate of the orthocentre is $-\dfrac{14}{5} = -2.8$.

Final Answer: -2.8

Question 21

English

Let L be the line $\dfrac{2x-1}{3} = \dfrac{3y+1}{2} = \dfrac{z-2}{-4}$. Then the number of points on L whose coordinates are all integers and whose z-coordinate satisfies $|z| \le 500$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना L रेखा $\dfrac{2x-1}{3} = \dfrac{3y+1}{2} = \dfrac{z-2}{-4}$ है। तब L पर स्थित उन बिन्दुओं की संख्या, जिनके सभी निर्देशांक पूर्णांक हैं

तथा जिनका z-निर्देशांक $|z| \leq 500$ को सन्तुष्ट करता है, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Put each ratio equal to t: $2x-1=3t$, $3y+1=2t$, $z-2=-4t$, so $x=\dfrac{1+3t}{2}$, $y=\dfrac{2t-1}{3}$, $z=2-4t$ (parametric form of L; it passes through $\left(\dfrac{1}{2},-\dfrac{1}{3},2\right)$ with direction ratios $\dfrac{3}{2},\dfrac{2}{3},-4$).

If x and y are integers then $3t=2x-1$ and $2t=3y+1$ are integers, so $t=3t-2t$ is an integer; then $z=2-4t$ is automatically an integer.

For integer t: $x\in\mathbb{Z}$ requires $1+3t$ even, that is t odd; $y\in\mathbb{Z}$ requires $3\mid 2t-1$, that is $t\equiv 2 \pmod 3$.

t odd and $t\equiv 2 \pmod 3$ together mean $t\equiv 5 \pmod 6$. Write $t=6k+5$, $k\in\mathbb{Z}$.

Then $x=\dfrac{16+18k}{2}=9k+8$, $y=\dfrac{12k+9}{3}=4k+3$, $z=2-24k-20=-24k-18$; every integer k gives an integer point, and distinct k give distinct points.

Condition $|z|\leq 500$: $-500\leq 24k+18\leq 500 \Rightarrow -518\leq 24k\leq 482 \Rightarrow -21.58\ldots\leq k\leq 20.08\ldots$

So $k=-21,-20,\ldots,20$: that is 42 values (end checks: $k=-21$ gives $z=486$, $k=20$ gives $z=-498$, $k=-22$ gives $z=510$, $k=21$ gives $z=-522$).

Final Answer: 42

Question 22

English

Let $\vec{a}$ and $\vec{b}$ be two vectors such that $|\vec{a}|=2$, $|\vec{b}|=5$ and the angle between $\vec{a}$ and $\vec{b}$ is $\dfrac{\pi}{3}$. Let $\vec{r}$ be a vector coplanar with $\vec{a}$ and $\vec{b}$ such that the scalar projection of $\vec{r}$ on $\vec{a}$ is 11 and the scalar projection of $\vec{r}$ on $\vec{b}$ is 13. Then the value of $|\vec{r}|^2$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $\vec{a}$ तथा $\vec{b}$ दो सदिश इस प्रकार हैं कि $|\vec{a}|=2$, $|\vec{b}|=5$ है तथा $\vec{a}$ और $\vec{b}$ के बीच का कोण $\dfrac{\pi}{3}$ है। माना $\vec{r}$, $\vec{a}$ तथा $\vec{b}$ के साथ समतलीय एक सदिश है जिसका $\vec{a}$ पर अदिश प्रक्षेप 11 तथा $\vec{b}$ पर अदिश प्रक्षेप 13 है। तब $|\vec{r}|^2$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The angle between $\vec{a}$ and $\vec{b}$ is $\dfrac{\pi}{3}$, so they are non-zero and non-collinear. By the linear combination theorem in a plane, every vector coplanar with them is $\vec{r}=x\vec{a}+y\vec{b}$ for unique real x, y.

$\vec{a}\cdot\vec{b}=|\vec{a}||\vec{b}|\cos\dfrac{\pi}{3}=5$, $\vec{a}\cdot\vec{a}=4$, $\vec{b}\cdot\vec{b}=25$.

The scalar projection of $\vec{r}$ on $\vec{a}$ is $\dfrac{\vec{r}\cdot\vec{a}}{|\vec{a}|}=11$, so $\vec{r}\cdot\vec{a}=22$; similarly $\vec{r}\cdot\vec{b}=13\cdot 5=65$.

$\vec{r}\cdot\vec{a}=4x+5y=22$ and $\vec{r}\cdot\vec{b}=5x+25y=65$, that is $x+5y=13$.

Subtracting: $3x=9$, so $x=3$ and $y=2$; $\vec{r}=3\vec{a}+2\vec{b}$.

$|\vec{r}|^2=9|\vec{a}|^2+4|\vec{b}|^2+12\,\vec{a}\cdot\vec{b}=36+100+60=196$.

(Equivalently $|\vec{r}|^2=\vec{r}\cdot\vec{r}=3(\vec{r}\cdot\vec{a})+2(\vec{r}\cdot\vec{b})=66+130=196$.)

Final Answer: 196

Question 23

English

Let $\vec{a}$ and $\vec{b}$ be two vectors of equal magnitude such that $\vec{a}+\vec{b} = 2\hat{i}+2\hat{j}+2\hat{k}$ and $\vec{a}\times\vec{b} = 6\hat{i}-6\hat{j}$. Then the value of $\vec{a}\cdot\vec{b}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $\vec{a}$ तथा $\vec{b}$ समान परिमाण वाले दो सदिश इस प्रकार हैं कि $\vec{a}+\vec{b} = 2\hat{i}+2\hat{j}+2\hat{k}$ तथा $\vec{a}\times\vec{b} = 6\hat{i}-6\hat{j}$ है। तब $\vec{a}\cdot\vec{b}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Put $\vec{s} = \vec{a}+\vec{b} = 2\hat{i}+2\hat{j}+2\hat{k}$ and $\vec{d} = \vec{a}-\vec{b}$. Since $|\vec{a}| = |\vec{b}|$, these are the diagonals of a rhombus.

Dot product: $\vec{s}\cdot\vec{d} = |\vec{a}|^2 - |\vec{b}|^2 = 0$, so the diagonals are perpendicular.

Cross product: $\vec{s}\times\vec{d} = \vec{a}\times\vec{a} - \vec{a}\times\vec{b} + \vec{b}\times\vec{a} - \vec{b}\times\vec{b} = -2\,\vec{a}\times\vec{b}$.

Taking magnitudes, with the angle between $\vec{s}$ and $\vec{d}$ equal to $\frac{\pi}{2}$: $|\vec{s}||\vec{d}| = 2|\vec{a}\times\vec{b}| = 2\sqrt{36+36} = 12\sqrt{2}$.

$|\vec{s}| = \sqrt{12} = 2\sqrt{3}$, so $|\vec{d}| = \dfrac{12\sqrt{2}}{2\sqrt{3}} = 2\sqrt{6}$ and $|\vec{d}|^2 = 24$.

Also $|\vec{s}|^2 - |\vec{d}|^2 = 4\,\vec{a}\cdot\vec{b}$, so $\vec{a}\cdot\vec{b} = \dfrac{12-24}{4} = -3$.

Check by finding the vectors: $\vec{d}\perp\vec{s}$ and $\vec{s}\times\vec{d} = -12\hat{i}+12\hat{j}$ give $\vec{d} = \dfrac{2(\vec{s}\times(\vec{a}\times\vec{b}))}{|\vec{s}|^2} = 2\hat{i}+2\hat{j}-4\hat{k}$, so $\vec{a} = 2\hat{i}+2\hat{j}-\hat{k}$, $\vec{b} = 3\hat{k}$; indeed $|\vec{a}| = |\vec{b}| = 3$, $\vec{a}\times\vec{b} = 6\hat{i}-6\hat{j}$ and $\vec{a}\cdot\vec{b} = -3$.

Final Answer: -3

Question 24

English

Let $OABC$ be a tetrahedron in which ABC is an equilateral triangle, $OA = OB = OC = 5$ and the perpendicular distance of O from the plane of ABC is 3. Let M be the mid-point of AC. If θ is the acute angle between the lines OA and BM, then the value of $\cos\theta$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $OABC$ एक चतुष्फलक है जिसमें ABC एक समबाहु त्रिभुज है, $OA = OB = OC = 5$ है तथा तल ABC से O की लम्बवत दूरी 3 है। माना M, AC का मध्यबिन्दु है। यदि रेखाओं OA तथा BM के बीच का न्यून कोण θ है, तो $\cos\theta$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let G be the foot of the perpendicular from O to the plane ABC. Then $GA^2 = OA^2 - OG^2 = 25 - 9 = 16$, and the same holds for GB, GC; so $GA = GB = GC = 4$ and G is the circumcentre of ABC, which for an equilateral triangle is also its centroid.

Choose G as the origin, the plane ABC as the xy-plane and $\overrightarrow{GA}$ along the x-axis. The vertices are at distance 4 from G and 120° apart: $A(4,0,0)$, $B(-2,2\sqrt{3},0)$, $C(-2,-2\sqrt{3},0)$, and $O(0,0,3)$.

Mid-point of AC: $M = (1,-\sqrt{3},0)$.

Direction vectors: $\overrightarrow{OA} = 4\hat{i}+0\hat{j}-3\hat{k}$ with $|\overrightarrow{OA}| = 5$; $\overrightarrow{BM} = 3\hat{i}-3\sqrt{3}\hat{j}+0\hat{k}$ with $|\overrightarrow{BM}| = \sqrt{9+27} = 6$.

$\overrightarrow{OA}\cdot\overrightarrow{BM} = 12+0+0 = 12$.

The lines OA and BM do not meet (they are skew), but the angle between two lines is the angle between their direction vectors, and the acute angle uses the absolute value: $\cos\theta = \dfrac{|\overrightarrow{OA}\cdot\overrightarrow{BM}|}{|\overrightarrow{OA}||\overrightarrow{BM}|} = \dfrac{12}{30} = 0.4$.

Final Answer: 0.4

Question 25

English

Let $\vec{a}$ and $\vec{b}$ be vectors such that $|\vec{a}| = 3$, $|\vec{a} + \vec{b}| = 4$ and $(2\vec{a} + \vec{b}) \times (\vec{a} - 2\vec{b}) = 20\hat{i} - 40\hat{j} + 40\hat{k}$. Then the value of $(\vec{a} + \vec{b}) \cdot (3\vec{a} - 2\vec{b})$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $\vec{a}$ तथा $\vec{b}$ ऐसे सदिश हैं कि $|\vec{a}| = 3$, $|\vec{a} + \vec{b}| = 4$ तथा $(2\vec{a} + \vec{b}) \times (\vec{a} - 2\vec{b}) = 20\hat{i} - 40\hat{j} + 40\hat{k}$ है। तब $(\vec{a} + \vec{b}) \cdot (3\vec{a} - 2\vec{b})$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Expand the cross product using $\vec{a} \times \vec{a} = \vec{b} \times \vec{b} = \vec{0}$ and $\vec{b} \times \vec{a} = -\vec{a} \times \vec{b}$:
$(2\vec{a} + \vec{b}) \times (\vec{a} - 2\vec{b}) = 2\vec{a} \times \vec{a} - 4\vec{a} \times \vec{b} + \vec{b} \times \vec{a} - 2\vec{b} \times \vec{b} = -5\,\vec{a} \times \vec{b}$.
Hence $5|\vec{a} \times \vec{b}| = |20\hat{i} - 40\hat{j} + 40\hat{k}| = \sqrt{400 + 1600 + 1600} = 60$, so $|\vec{a} \times \vec{b}| = 12$.
Let $p = \vec{a} \cdot \vec{b}$ and $s = |\vec{b}|^2$. Lagrange's identity $|\vec{a} \times \vec{b}|^2 = |\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2$ gives $144 = 9s - p^2$.
From $|\vec{a} + \vec{b}|^2 = |\vec{a}|^2 + 2\vec{a} \cdot \vec{b} + |\vec{b}|^2$: $16 = 9 + 2p + s$, so $s = 7 - 2p$.
Substituting: $9(7 - 2p) - p^2 = 144 \Rightarrow p^2 + 18p + 81 = 0 \Rightarrow (p + 9)^2 = 0$.
The double root gives $p = -9$ uniquely, and then $s = 7 + 18 = 25$, that is $|\vec{b}| = 5$ (consistent: $9 \cdot 25 - 81 = 144$).
Now $(\vec{a} + \vec{b}) \cdot (3\vec{a} - 2\vec{b}) = 3|\vec{a}|^2 - 2\vec{a} \cdot \vec{b} + 3\vec{b} \cdot \vec{a} - 2|\vec{b}|^2 = 3|\vec{a}|^2 + \vec{a} \cdot \vec{b} - 2|\vec{b}|^2$.
$= 27 + (-9) - 50 = -32$.

Final Answer: -32

Question 26

English

Let $\vec{a}$ and $\vec{b}$ be two non-collinear vectors, and let P, Q and S be the points with position vectors $3\vec{a} - \vec{b}$, $\vec{a} + \vec{b}$ and $-\vec{a} + 2\vec{b}$, respectively, with respect to the origin O. The line through S parallel to the vector $\vec{a} + \vec{b}$ meets the line PQ at the point R. If $\overrightarrow{OR} = \lambda\vec{a} + \mu\vec{b}$, where λ and μ are real numbers, then the value of $\lambda\mu$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $\vec{a}$ तथा $\vec{b}$ दो असरेख सदिश हैं, तथा मूलबिन्दु O के सापेक्ष P, Q और S वे बिन्दु हैं जिनके स्थिति सदिश क्रमशः $3\vec{a} - \vec{b}$, $\vec{a} + \vec{b}$ तथा $-\vec{a} + 2\vec{b}$ हैं। S से होकर सदिश $\vec{a} + \vec{b}$ के समान्तर रेखा, रेखा PQ से बिन्दु R पर मिलती है। यदि $\overrightarrow{OR} = \lambda\vec{a} + \mu\vec{b}$ है, जहाँ λ तथा μ वास्तविक संख्याएँ हैं, तो $\lambda\mu$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$\overrightarrow{PQ} = (\vec{a} + \vec{b}) - (3\vec{a} - \vec{b}) = -2\vec{a} + 2\vec{b}$, which is not parallel to $\vec{a} + \vec{b}$, so the two lines meet in exactly one point.

A point of line PQ: $(3\vec{a} - \vec{b}) + s(-2\vec{a} + 2\vec{b}) = (3 - 2s)\vec{a} + (2s - 1)\vec{b}$, $s \in \mathbb{R}$.

A point of the line through S parallel to $\vec{a} + \vec{b}$: $(-\vec{a} + 2\vec{b}) + t(\vec{a} + \vec{b}) = (t - 1)\vec{a} + (t + 2)\vec{b}$, $t \in \mathbb{R}$.

At R the two expressions are equal. Since $\vec{a}$ and $\vec{b}$ are non-collinear, $x\vec{a} + y\vec{b} = \vec{0}$ only for $x = y = 0$, so the coefficients of $\vec{a}$ and of $\vec{b}$ must be equal separately:

$3 - 2s = t - 1$ and $2s - 1 = t + 2$.

Adding the two equations: $2 = 2t + 1$, so $t = \frac{1}{2}$; then $2s = t + 3 = \frac{7}{2}$, $s = \frac{7}{4}$.

Hence $\overrightarrow{OR} = -\frac{1}{2}\vec{a} + \frac{5}{2}\vec{b}$ (check with $s = \frac{7}{4}$: $3 - \frac{7}{2} = -\frac{1}{2}$, $\frac{7}{2} - 1 = \frac{5}{2}$).

By uniqueness of the coefficients, $\lambda = -\frac{1}{2}$ and $\mu = \frac{5}{2}$, so $\lambda\mu = -\frac{5}{4} = -1.25$.

Final Answer: -1.25

Question 27

English

Let $\vec{a}$ and $\vec{b}$ be vectors such that $|\vec{a}| = 2$, $|\vec{b}| = 3$ and $\vec{a} \cdot \vec{b} = 4$. Define the vectors $\vec{c}_1, \vec{c}_2, \vec{c}_3, \ldots$ by $\vec{c}_1 = \vec{b}$ and $\vec{c}_{n+1} = \vec{a} \times \vec{c}_n$ for every integer $n \geq 1$. Then the value of $|\vec{c}_2 + \vec{c}_3 + \vec{c}_4 + \vec{c}_5|^2$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $\vec{a}$ तथा $\vec{b}$ ऐसे सदिश हैं कि $|\vec{a}| = 2$, $|\vec{b}| = 3$ तथा $\vec{a} \cdot \vec{b} = 4$ है। सदिशों $\vec{c}_1, \vec{c}_2, \vec{c}_3, \ldots$ को $\vec{c}_1 = \vec{b}$ तथा प्रत्येक पूर्णांक $n \geq 1$ के लिए $\vec{c}_{n+1} = \vec{a} \times \vec{c}_n$ द्वारा परिभाषित किया जाता है। तब $|\vec{c}_2 + \vec{c}_3 + \vec{c}_4 + \vec{c}_5|^2$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$\vec{c}_2 = \vec{a} \times \vec{c}_1 = \vec{a} \times \vec{b}$. By Lagrange's identity, $|\vec{a} \times \vec{b}|^2 = |\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2 = 4 \cdot 9 - 16 = 20$.
Every $\vec{c}_n$ with $n \geq 2$ is a cross product with $\vec{a}$ as the first factor, so $\vec{c}_n \perp \vec{a}$ for $n \geq 2$.
For a vector $\vec{v} \perp \vec{a}$: $\vec{a} \times (\vec{a} \times \vec{v}) = (\vec{a} \cdot \vec{v})\vec{a} - (\vec{a} \cdot \vec{a})\vec{v} = -4\vec{v}$.
Taking $\vec{v} = \vec{c}_2$ and $\vec{v} = \vec{c}_3$: $\vec{c}_4 = -4\vec{c}_2$ and $\vec{c}_5 = -4\vec{c}_3$.
So $\vec{c}_2 + \vec{c}_3 + \vec{c}_4 + \vec{c}_5 = -3\vec{c}_2 - 3\vec{c}_3 = -3(\vec{c}_2 + \vec{c}_3)$.
$\vec{c}_3 = \vec{a} \times \vec{c}_2$ is perpendicular to $\vec{c}_2$, so $\vec{c}_2 \cdot \vec{c}_3 = 0$.
Since $\vec{c}_2 \perp \vec{a}$, the angle between them is $\frac{\pi}{2}$ and $|\vec{c}_3| = |\vec{a}||\vec{c}_2| \sin\frac{\pi}{2} = 2\sqrt{20}$, so $|\vec{c}_3|^2 = 80$.
Therefore $|-3(\vec{c}_2 + \vec{c}_3)|^2 = 9\left(|\vec{c}_2|^2 + 2\,\vec{c}_2 \cdot \vec{c}_3 + |\vec{c}_3|^2\right) = 9(20 + 0 + 80) = 900$.

Final Answer: 900

Question 28

English

Let $A(2,1,0)$, $B(-3,5,0)$ and $C(3,-3,c)$, where $c > 0$, be the vertices of a triangle such that the area of triangle ABC is $\frac{25}{16}$ times the area of the triangle whose vertices are the feet of the perpendiculars drawn from A, B and C to the xy-plane. Let θ be the acute angle between the plane of triangle ABC and the yz-plane. Then the value of $\cos\theta$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $A(2,1,0)$, $B(-3,5,0)$ तथा $C(3,-3,c)$, जहाँ $c > 0$ है, एक त्रिभुज के शीर्ष हैं जिससे त्रिभुज ABC का क्षेत्रफल उस त्रिभुज के क्षेत्रफल का $\frac{25}{16}$ गुना है जिसके शीर्ष A, B तथा C से xy-तल पर डाले गए लम्बों के पाद हैं। माना कि θ त्रिभुज ABC के तल तथा yz-तल के बीच का न्यून कोण है। तब $\cos\theta$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The feet of the perpendiculars from A, B, C to the xy-plane are $A'(2,1,0)$, $B'(-3,5,0)$, $C'(3,-3,0)$; the third coordinate is dropped and the first two are unchanged.

Area of $A'B'C'$ by the coordinate formula $\frac{1}{2}\,|x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)|$:

$\frac{1}{2}\,|2(5-(-3))+(-3)(-3-1)+3(1-5)| = \frac{1}{2}|16+12-12| = 8.$

Hence the area of triangle ABC is $\frac{25}{16}\times 8 = \frac{25}{2}$.

$\overrightarrow{AB} = (-5,4,0)$ and $\overrightarrow{AC} = (1,-4,c)$, so $\overrightarrow{AB}\times\overrightarrow{AC} = (4c-0,\ 0-(-5c),\ 20-4) = (4c,\ 5c,\ 16)$.

Area of $ABC = \frac{1}{2}|\overrightarrow{AB}\times\overrightarrow{AC}| = \frac{1}{2}\sqrt{16c^2+25c^2+256}$.

$\frac{1}{2}\sqrt{41c^2+256} = \frac{25}{2} \Rightarrow 41c^2+256 = 625 \Rightarrow c^2 = 9$, and $c > 0$ gives $c = 3$.

Then a normal to the plane ABC is $\vec{n} = (12,15,16)$ with $|\vec{n}| = \sqrt{144+225+256} = 25$. (Check: the $\hat{k}$-component 16 gives cos of the angle with the xy-plane $= \frac{16}{25}$, matching the area ratio.)

The angle between two planes is the angle between their normals; the yz-plane has normal $\hat{i}$.

$\cos\theta = \frac{|\vec{n}\cdot\hat{i}|}{|\vec{n}|} = \frac{12}{25} = 0.48.$

Final Answer: 0.48

Question 29

English

Let $\vec{a}$ and $\vec{b}$ be two vectors with $|\vec{a}| = |\vec{b}| = 5$ such that $\vec{a} \cdot \vec{b} - |\vec{a} \times \vec{b}| = 17$. Let O be the origin and let A and B be the points such that $\overrightarrow{OA} = \vec{a}$ and $\overrightarrow{OB} = \vec{b}$. Then the area of triangle OAB is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $\vec{a}$ तथा $\vec{b}$ दो सदिश हैं जिनके लिए $|\vec{a}| = |\vec{b}| = 5$ तथा $\vec{a} \cdot \vec{b} - |\vec{a} \times \vec{b}| = 17$ है। माना कि O मूलबिन्दु है तथा A और B ऐसे बिन्दु हैं कि $\overrightarrow{OA} = \vec{a}$ तथा $\overrightarrow{OB} = \vec{b}$ है। तब त्रिभुज OAB का क्षेत्रफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $p = \vec{a} \cdot \vec{b}$ and $q = |\vec{a} \times \vec{b}|$; note $q \geq 0$ because it is a magnitude.

Lagrange's identity: $|\vec{a} \times \vec{b}|^2 + (\vec{a} \cdot \vec{b})^2 = |\vec{a}|^2 |\vec{b}|^2$, so $p^2 + q^2 = 25 \cdot 25 = 625$.

The given condition is $p = q + 17$. Substituting: $(q+17)^2 + q^2 = 625 \Rightarrow 2q^2 + 34q + 289 - 625 = 0 \Rightarrow q^2 + 17q - 168 = 0$.

Factorising: $(q-7)(q+24) = 0$, so $q = 7$ or $q = -24$; the second is rejected since $q \geq 0$.

Hence $q = 7$ and $p = 24$ (check: $24^2 + 7^2 = 576 + 49 = 625$).

The area of triangle OAB is $\dfrac{1}{2}|\vec{a} \times \vec{b}| = \dfrac{1}{2} \cdot 7 = 3.5$.

Final Answer: 3.5

Question 30

English

Let $\vec{a}$ and $\vec{b}$ be vectors with $|\vec{a}| = 2$ and $|\vec{b}| = 3$, where the angle between $\vec{a}$ and $\vec{b}$ may take any value in $[0, \pi]$. Then the maximum value of $|\vec{a} \times \vec{b}|^2 + 2|\vec{a} + \vec{b}|^2$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $\vec{a}$ तथा $\vec{b}$ ऐसे सदिश हैं कि $|\vec{a}| = 2$ तथा $|\vec{b}| = 3$ है, जहाँ $\vec{a}$ तथा $\vec{b}$ के बीच का कोण $[0, \pi]$ में कोई भी मान ले सकता है। तब $|\vec{a} \times \vec{b}|^2 + 2|\vec{a} + \vec{b}|^2$ का अधिकतम मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $p = \vec{a} \cdot \vec{b} = 6\cos\theta$, where $\theta \in [0, \pi]$ is the angle between the vectors, so p takes every value in $[-6, 6]$.

Lagrange's identity: $|\vec{a} \times \vec{b}|^2 = |\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2 = 36 - p^2$.

Dot product expansion: $|\vec{a} + \vec{b}|^2 = |\vec{a}|^2 + 2\vec{a} \cdot \vec{b} + |\vec{b}|^2 = 13 + 2p$.

So the expression is $E(p) = 36 - p^2 + 2(13 + 2p) = 62 + 4p - p^2 = 66 - (p-2)^2$.

E is largest when $(p-2)^2$ is smallest; $p = 2$ lies in $[-6, 6]$ (it corresponds to $\cos\theta = \dfrac{1}{3}$).

Hence the maximum value is $E(2) = 66$.

Final Answer: 66

Question 31

English

Let $L_1 : \dfrac{x+1}{2} = \dfrac{y+1}{3} = \dfrac{z+2}{2}$ and $L_2 : \dfrac{x-6}{1} = \dfrac{y}{2} = \dfrac{z-4}{2}$. Let d be the shortest distance between L_1 and L_2, and let the line of shortest distance between L_1 and L_2 meet the plane $x = 0$ at the point $(0, \beta, \gamma)$. Then the value of $d + \beta + \gamma$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $L_1 : \dfrac{x+1}{2} = \dfrac{y+1}{3} = \dfrac{z+2}{2}$ तथा $L_2 : \dfrac{x-6}{1} = \dfrac{y}{2} = \dfrac{z-4}{2}$ हैं। माना कि L_1 तथा L_2 के बीच की न्यूनतम दूरी d है, तथा L_1 और L_2 के बीच की न्यूनतम दूरी की रेखा, तल $x = 0$ से बिन्दु $(0, \beta, \gamma)$ पर मिलती है। तब $d + \beta + \gamma$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

General points: $P(-1 + 2s,\ -1 + 3s,\ -2 + 2s)$ on L_1 and $Q(6 + t,\ 2t,\ 4 + 2t)$ on L_2.

$\overrightarrow{PQ} = (7 + t - 2s,\ 1 + 2t - 3s,\ 6 + 2t - 2s)$.

For the line of shortest distance, $\overrightarrow{PQ}$ is perpendicular to both directions $(2, 3, 2)$ and $(1, 2, 2)$:

$2(7 + t - 2s) + 3(1 + 2t - 3s) + 2(6 + 2t - 2s) = 29 + 12t - 17s = 0,$

$(7 + t - 2s) + 2(1 + 2t - 3s) + 2(6 + 2t - 2s) = 21 + 9t - 12s = 0.$

From the second, $t = \dfrac{4s - 7}{3}$; substituting in the first: $29 + 4(4s - 7) - 17s = 1 - s = 0$, so $s = 1$ and $t = -1$.

Hence $P(1, 2, 0)$, $Q(5, -2, 2)$ and $\overrightarrow{PQ} = (4, -4, 2)$, so $d = PQ = \sqrt{16 + 16 + 4} = 6$.

Check by the formula: $(2, 3, 2) \times (1, 2, 2) = (2, -2, 1)$ and $\dfrac{|(6 - (-1), 0 - (-1), 4 - (-2)) \cdot (2, -2, 1)|}{3} = \dfrac{|14 - 2 + 6|}{3} = 6.$

The line of shortest distance passes through $P(1, 2, 0)$ with direction ratios $2, -2, 1$: $\dfrac{x-1}{2} = \dfrac{y-2}{-2} = \dfrac{z}{1} = k.$

On $x = 0$: $k = -\dfrac{1}{2}$, so $y = 2 + 1 = 3$ and $z = -\dfrac{1}{2}$. Thus $\beta = 3$, $\gamma = -\dfrac{1}{2}$.

$d + \beta + \gamma = 6 + 3 - \dfrac{1}{2} = 8.5.$

Final Answer: 8.5

Question 32

English

Let L_1 be the line $\frac{x-1}{2} = \frac{y+1}{-1} = \frac{z-a}{1}$, where a is a negative real number. Let L_2 be the line through the point $(2, \mu, 0)$ parallel to the vector $\hat{i} + 2\hat{j} - 2\hat{k}$, where μ is the real number for which L_2 intersects the line $L_3 : \frac{x}{1} = \frac{y+1}{2} = \frac{z+1}{1}$. The shortest distance between L_1 and L_2 is $3\sqrt{2}$. If the line of shortest distance between L_1 and L_2 meets L_1 at the point (α, β, γ), then the value of $\alpha + \beta + \gamma$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि L_1 रेखा $\frac{x-1}{2} = \frac{y+1}{-1} = \frac{z-a}{1}$ है, जहाँ a एक ऋणात्मक वास्तविक संख्या है। माना कि L_2 बिन्दु $(2, \mu, 0)$ से होकर जाने वाली तथा सदिश $\hat{i} + 2\hat{j} - 2\hat{k}$ के समान्तर रेखा है, जहाँ μ वह वास्तविक संख्या है जिसके लिए L_2 रेखा $L_3 : \frac{x}{1} = \frac{y+1}{2} = \frac{z+1}{1}$ को प्रतिच्छेद करती है। L_1 तथा L_2 के बीच की न्यूनतम दूरी $3\sqrt{2}$ है। यदि L_1 तथा L_2 के बीच की न्यूनतम दूरी की रेखा L_1 से बिन्दु (α, β, γ) पर मिलती है, तो $\alpha + \beta + \gamma$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Step 1 (μ). L_2 passes through $B(2, \mu, 0)$ with direction $\vec{v} = (1, 2, -2)$; L_3 passes through $C(0, -1, -1)$ with direction $\vec{w} = (1, 2, 1)$. These directions are not parallel, so the lines intersect if and only if they are coplanar:

$\begin{vmatrix} 2-0 & \mu+1 & 0+1 \\ 1 & 2 & -2 \\ 1 & 2 & 1 \end{vmatrix} = 2(2+4) - (\mu+1)(1+2) + 1(2-2) = 12 - 3(\mu+1) = 0$, so $\mu = 3$ and $B = (2, 3, 0)$.

Step 2 (a). L_1 passes through $A(1, -1, a)$ with direction $\vec{u} = (2, -1, 1)$. $\vec{u} \times \vec{v} = ((-1)(-2) - 1 \cdot 2,\ 1 \cdot 1 - 2(-2),\ 2 \cdot 2 - (-1) \cdot 1) = (0, 5, 5)$, of magnitude $5\sqrt{2}$.

The scalar triple product $[\overrightarrow{AB}\ \vec{u}\ \vec{v}] = (1, 4, -a) \cdot (0, 5, 5) = 20 - 5a$.

Shortest distance $= \frac{|20 - 5a|}{5\sqrt{2}} = 3\sqrt{2} \Rightarrow |20 - 5a| = 30 \Rightarrow a = -2$ or $a = 10$; since $a < 0$, $a = -2$ and $A = (1, -1, -2)$.

Step 3 (line of shortest distance). General points: $P(1 + 2s,\ -1 - s,\ -2 + s)$ on L_1 and $Q(2 + t,\ 3 + 2t,\ -2t)$ on L_2.

$\overrightarrow{PQ} = (1 + t - 2s,\ 4 + 2t + s,\ 2 - 2t - s)$ must be parallel to $\vec{u} \times \vec{v} = (0, 5, 5)$, that is, first component 0 and second = third:

$1 + t - 2s = 0$ and $4 + 2t + s = 2 - 2t - s \Rightarrow 2t + s = -1$.

With $t = 2s - 1$: $4s - 2 + s = -1 \Rightarrow s = \frac{1}{5}$, $t = -\frac{3}{5}$.

So $P = \left(\frac{7}{5}, -\frac{6}{5}, -\frac{9}{5}\right)$ and $Q = \left(\frac{7}{5}, \frac{9}{5}, \frac{6}{5}\right)$; $\overrightarrow{PQ} = (0, 3, 3)$ has length $3\sqrt{2}$, as it must.

$\alpha + \beta + \gamma = \frac{7}{5} - \frac{6}{5} - \frac{9}{5} = -\frac{8}{5} = -1.6$.

Final Answer: -1.6

Question 33

English

Let $L_1 : \dfrac{x-1}{2} = \dfrac{y-2}{1} = \dfrac{z-4}{1}$ and $L_2 : \dfrac{x-2}{1} = \dfrac{y-\mu}{-1} = \dfrac{z}{2}$, where μ is a real number such that L_1 and L_2 are skew lines and the line of shortest distance between L_1 and L_2 intersects the z-axis. Let d be the shortest distance between L_1 and L_2 for this value of μ. Then the value of $\mu - d^2$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $L_1 : \dfrac{x-1}{2} = \dfrac{y-2}{1} = \dfrac{z-4}{1}$ तथा $L_2 : \dfrac{x-2}{1} = \dfrac{y-\mu}{-1} = \dfrac{z}{2}$ हैं, जहाँ μ एक ऐसी वास्तविक संख्या है कि L_1 तथा L_2 विषमतलीय रेखाएँ हैं तथा L_1 और L_2 के बीच की न्यूनतम दूरी की रेखा z-अक्ष को प्रतिच्छेद करती है। माना कि μ के इस मान के लिए L_1 तथा L_2 के बीच की न्यूनतम दूरी d है। तब $\mu - d^2$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

General points: $P(1+2s,\ 2+s,\ 4+s)$ on L_1 and $Q(2+t,\ \mu-t,\ 2t)$ on L_2; $\overrightarrow{PQ} = (1+t-2s,\ \mu-2-t-s,\ 2t-4-s)$.

$\overrightarrow{PQ} \perp (2,1,1)$: $2(1+t-2s)+(\mu-2-t-s)+(2t-4-s) = 3t-6s+\mu-4 = 0$.

$\overrightarrow{PQ} \perp (1,-1,2)$: $(1+t-2s)-(\mu-2-t-s)+2(2t-4-s) = 6t-3s-\mu-5 = 0$.

Twice the first minus the second: $-9s+3\mu-3=0$, so $s = \dfrac{\mu-1}{3}$; then $3t = 6s-\mu+4 = \mu+2$, so $t = \dfrac{\mu+2}{3}$.

Hence $P = \left(\dfrac{2\mu+1}{3}, \dfrac{\mu+5}{3}, \dfrac{\mu+11}{3}\right)$, and the line of shortest distance has direction $(2,1,1)\times(1,-1,2) = (3,-3,-3)$, that is $(1,-1,-1)$.

The z-axis passes through O with direction $(0,0,1)$. It is not parallel to $(1,-1,-1)$, so the two lines intersect if and only if they are coplanar:

$$\begin{vmatrix} x_P-0 & y_P-0 & z_P-0 \\ 0 & 0 & 1 \\ 1 & -1 & -1 \end{vmatrix} = x_P(0+1) - y_P(0-1) + z_P(0) = x_P + y_P = 0.$$

$$\frac{2\mu+1}{3} + \frac{\mu+5}{3} = \frac{3\mu+6}{3} = 0 \Rightarrow \mu = -2.$$

For $\mu=-2$: $s=-1$, $t=0$, $P(-1,1,3)$, $Q(2,-2,0)$, and $\overrightarrow{PQ} = (3,-3,-3)$, so $d = 3\sqrt{3} \neq 0$ and the lines are indeed skew. (The line PQ meets the z-axis at $(0,0,2)$.)

Check: $\dfrac{|(2-1,-2-2,0-4)\cdot(3,-3,-3)|}{|(3,-3,-3)|} = \dfrac{|3+12+12|}{3\sqrt{3}} = 3\sqrt{3}$.

$\mu - d^2 = -2 - 27 = -29$.

Final Answer: -29

Question 34

English

Let L_1 be the line of intersection of the planes $x + y = 3$ and $y - z = 1$, and let L_2 be the line passing through the points $(3, 4, -3)$ and $(5, 4, 1)$. The line of shortest distance between L_1 and L_2 meets L_1 at the point P and L_2 at the point Q. If O is the origin, then the value of $OP^2 + OQ^2$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि L_1 तलों $x + y = 3$ तथा $y - z = 1$ की प्रतिच्छेदन रेखा है, तथा L_2 बिन्दुओं $(3, 4, -3)$ और $(5, 4, 1)$ से होकर जाने वाली रेखा है। L_1 तथा L_2 के बीच की न्यूनतम दूरी की रेखा L_1 से बिन्दु P पर तथा L_2 से बिन्दु Q पर मिलती है। यदि O मूलबिन्दु है, तो $OP^2 + OQ^2$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

L_1 lies in both planes, so its direction is perpendicular to both normals: $(1, 1, 0) \times (0, 1, -1) = (1 \cdot (-1) - 0 \cdot 1,\ 0 \cdot 0 - 1 \cdot (-1),\ 1 \cdot 1 - 1 \cdot 0) = (-1, 1, 1)$.

Putting $y = 0$ in the two equations gives $x = 3$, $z = -1$, so $(3, 0, -1) \in L_1$ and a general point is $P(3 - t,\ t,\ t - 1)$.

L_2 has direction $(5-3, 4-4, 1+3) = (2, 0, 4)$, that is $(1, 0, 2)$, and a general point is $Q(3+s,\ 4,\ -3+2s)$.

$\overrightarrow{PQ} = (s + t,\ 4 - t,\ 2s - t - 2)$.

$\overrightarrow{PQ} \perp (-1, 1, 1)$: $-(s + t) + (4 - t) + (2s - t - 2) = s - 3t + 2 = 0$.

$\overrightarrow{PQ} \perp (1, 0, 2)$: $(s + t) + 2(2s - t - 2) = 5s - t - 4 = 0$.

From the first, $s = 3t - 2$; then $15t - 10 - t - 4 = 0 \Rightarrow t = 1,\ s = 1$.

So $P = (2, 1, 0)$ and $Q = (4, 4, -1)$; $\overrightarrow{PQ} = (2, 3, -1)$, and $(-1, 1, 1) \times (1, 0, 2) = (2, 3, -1)$, as expected. The line of shortest distance is $\dfrac{x-2}{2} = \dfrac{y-1}{3} = \dfrac{z}{-1}$.

$OP^2 = 4 + 1 + 0 = 5$ and $OQ^2 = 16 + 16 + 1 = 33$, so $OP^2 + OQ^2 = 38$.

Final Answer: 38

Question 35

English

Let Π be the plane $2x-y+2z=4$ and let ℓ be the line of intersection of Π and the plane $x+2z=6$. Let A be the point $(3,-1,3)$, and let B and C be the two distinct points on ℓ such that each of the lines AB and AC makes an angle $\sin^{-1}\left(\frac{3}{5}\right)$ with the plane Π. Then the value of $\cos\angle BAC$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि Π तल $2x-y+2z=4$ है तथा ℓ, तल Π और तल $x+2z=6$ की प्रतिच्छेदन रेखा है। माना कि A बिन्दु $(3,-1,3)$ है, तथा B और C रेखा ℓ पर ऐसे दो भिन्न बिन्दु हैं कि रेखाओं AB तथा AC में से प्रत्येक तल Π के साथ $\sin^{-1}\left(\frac{3}{5}\right)$ का कोण बनाती है। तब $\cos\angle BAC$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The normals of Π and of $x+2z=6$ are $(2,-1,2)$ and $(1,0,2)$. The direction of their line of intersection is $(2,-1,2)\times(1,0,2)=(-2\cdot 2-2\cdot 0,\ 2\cdot 1-2\cdot 2,\ 2\cdot 0-(-1)\cdot 1)=(-2,-2,1)$.

A common point: $x=0$ gives $z=3$ from $x+2z=6$, and then $-y+6=4$ gives $y=2$. So $(0,2,3)$ lies on ℓ and $\ell:\ (x,y,z)=(-2t,\ 2-2t,\ 3+t),\ t\in\mathbb{R}$.

Distance of $A(3,-1,3)$ from Π: $h=\dfrac{|2(3)-(-1)+2(3)-4|}{\sqrt{4+1+4}}=\dfrac{9}{3}=3$.

If a line through A meets Π at P and makes an angle θ with Π, then $h=AP\sin\theta$ (the perpendicular from A, the segment AP and its projection on Π form a right triangle). Every point P of ℓ lies on Π, so $\sin\theta=\dfrac{3}{AP}$.

The condition $\sin\theta=\dfrac{3}{5}$ is therefore $AP=5$.

For $P(-2t,\ 2-2t,\ 3+t)$: $\overrightarrow{AP}=(-2t-3,\ 3-2t,\ t)$ and $AP^2=(2t+3)^2+(3-2t)^2+t^2=8t^2+18+t^2=9t^2+18$.

$9t^2+18=25\Rightarrow t^2=\dfrac{7}{9}\Rightarrow t=\pm\dfrac{\sqrt{7}}{3}$. Two distinct points, so these are B and C, and $AB=AC=5$.

With $\overrightarrow{AB}=(-2t-3,\ 3-2t,\ t)$ and $\overrightarrow{AC}=(2t-3,\ 3+2t,\ -t)$ for $t=\dfrac{\sqrt{7}}{3}$:

$\overrightarrow{AB}\cdot\overrightarrow{AC}=-(2t+3)(2t-3)+(3-2t)(3+2t)-t^2=(9-4t^2)+(9-4t^2)-t^2=18-9t^2=18-7=11$.

Hence $\cos\angle BAC=\dfrac{\overrightarrow{AB}\cdot\overrightarrow{AC}}{AB\cdot AC}=\dfrac{11}{25}=0.44$.

Final Answer: 0.44

Question 36

English

Let P_1 and P_2 be the planes $x + 2y + 2z = 5$ and $11x + 10y - 2z = 7$, respectively. Let d_1 and d_2 be the distances of the point $A(2, 0, -9)$ from the two planes that bisect the angles between P_1 and P_2. Then the value of $d_1^2 + d_2^2$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि P_1 तथा P_2 क्रमशः तल $x+2y+2z = 5$ तथा $11x+10y-2z = 7$ हैं। माना कि d_1 तथा d_2 बिन्दु $A(2, 0, -9)$ की उन दो तलों से दूरियाँ हैं जो P_1 तथा P_2 के बीच के कोणों को समद्विभाजित करते हैं। तब $d_1^2 + d_2^2$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$|(1, 2, 2)| = 3$ and $|(11, 10, -2)| = \sqrt{121 + 100 + 4} = 15$. A point is on a bisecting plane exactly when its distances from P_1 and P_2 are equal:

$$\frac{x + 2y + 2z - 5}{3} = \pm\frac{11x + 10y - 2z - 7}{15} \iff 5(x + 2y + 2z - 5) = \pm(11x + 10y - 2z - 7).$$

Sign $+$: $5x + 10y + 10z - 25 - 11x - 10y + 2z + 7 = 0 \Rightarrow -6x + 12z - 18 = 0 \Rightarrow x - 2z + 3 = 0$.

Sign $-$: $5x+10y+10z-25+11x+10y-2z-7 = 0 \Rightarrow 16x+20y+8z-32 = 0 \Rightarrow 4x+5y+2z-8 = 0$.

(The normals $(1, 0, -2)$ and $(4, 5, 2)$ satisfy $4+0-4 = 0$: the two bisecting planes are perpendicular, as they must be.)

Distance of $A(2, 0, -9)$ from $x - 2z + 3 = 0$: $d_1 = \dfrac{|2 + 18 + 3|}{\sqrt{5}} = \dfrac{23}{\sqrt{5}}$, so $d_1^2 = \dfrac{529}{5}$.

Distance of A from $4x + 5y + 2z - 8 = 0$: $d_2 = \dfrac{|8 + 0 - 18 - 8|}{\sqrt{45}} = \dfrac{18}{\sqrt{45}}$, so $d_2^2 = \dfrac{324}{45} = \dfrac{36}{5}$.

$d_1^2 + d_2^2 = \dfrac{529 + 36}{5} = \dfrac{565}{5} = 113$.

Check: both bisecting planes contain the line $P_1 \cap P_2$, which passes through $(1, 0, 2)$ with direction $(1, 2, 2) \times (11, 10, -2) = (-24, 24, -12) \parallel (2, -2, 1)$. As the planes are perpendicular, $d_1^2 + d_2^2$ is the square of the distance of A from this line: $\overrightarrow{v} = (1, 0, -11)$, $|\overrightarrow{v}|^2 = 122$, $\left(\dfrac{\overrightarrow{v} \cdot (2, -2, 1)}{3}\right)^2 = \left(\dfrac{-9}{3}\right)^2 = 9$, and $122 - 9 = 113$.

Final Answer: 113

Question 37

English

Let Π be the plane $2x - y + 2z = d$, where d is a real number, such that the image of the point $A(1, 2, 3)$ in Π lies on the line $L : \ x + 2 = y - 5 = z$. Let Q be the point of intersection of L and Π. Then the value of $(AQ)^2$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि Π तल $2x - y + 2z = d$ है, जहाँ d एक वास्तविक संख्या है, इस प्रकार कि तल Π में बिन्दु $A(1, 2, 3)$ का प्रतिबिम्ब रेखा $L : \ x + 2 = y - 5 = z$ पर स्थित है। माना कि Q, रेखा L तथा तल Π का प्रतिच्छेदन बिन्दु है। तब $(AQ)^2$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The image A' of A in the plane $2x-y+2z = d$ is given by $\frac{x-1}{2} = \frac{y-2}{-1} = \frac{z-3}{2} = -\frac{2(2 \cdot 1 - 2 + 2 \cdot 3 - d)}{4+1+4} = -\frac{2(6-d)}{9} = s$, say.

So $A' = (1 + 2s, \ 2 - s, \ 3 + 2s)$.

L is $x + 2 = y - 5 = z$. For A': $x + 2 = 3 + 2s$, $y - 5 = -3 - s$, $z = 3 + 2s$.

The first and third are always equal; equating the first two: $3 + 2s = -3 - s \Rightarrow 3s = -6 \Rightarrow s = -2$. Then $A' = (-3, 4, -1)$, and indeed $-3 + 2 = 4 - 5 = -1$.

From $s = -\frac{2(6-d)}{9} = -2$: $6 - d = 9 \Rightarrow d = -3$, so $\Pi : \ 2x - y + 2z = -3$. (Check: the midpoint of AA' is $(-1, 3, 1)$ and $-2 - 3 + 2 = -3$.)

A general point of L is $(t-2, \ t+5, \ t)$. It lies on Π when $2(t-2)-(t+5)+2t = -3$, i.e. $3t-9 = -3$, so $t = 2$ and $Q = (0, 7, 2)$. (L is not parallel to Π, since $(1, 1, 1) \cdot (2, -1, 2) = 3 \neq 0$.)

$\overrightarrow{AQ} = (-1, 5, -1)$, so $(AQ)^2 = 1 + 25 + 1 = 27$.

Final Answer: 27

Question 38

English

For each real number λ, let Π_λ be the plane $(4x - 3z - 7) + \lambda(4x + y - 3z - 9) = 0$, and let A_λ denote the image of the point $A(2, 2, -2)$ in the plane Π_λ. Then the maximum value of the distance between A_λ and A_μ, as λ and μ vary over $\mathbb{R}$, is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

प्रत्येक वास्तविक संख्या λ के लिए, माना कि Π_λ तल $(4x - 3z - 7) + \lambda(4x + y - 3z - 9) = 0$ है, तथा A_λ तल Π_λ में बिन्दु $A(2, 2, -2)$ का प्रतिबिम्ब है। जब λ तथा μ सभी वास्तविक मानों पर परिवर्तित होते हैं, तब A_λ तथा A_μ के बीच की दूरी का अधिकतम मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Π_λ is the family of planes through the line L of intersection of $4x-3z-7=0$ and $4x+y-3z-9=0$; every member contains L, and the members are all planes through L except $4x + y - 3z = 9$ itself.

Direction of L: $(4, 0, -3) \times (4, 1, -3) = (0 \cdot (-3) - (-3) \cdot 1,\ (-3) \cdot 4 - 4 \cdot (-3),\ 4 \cdot 1 - 0 \cdot 4) = (3, 0, 4)$. The point $P_0(1, 2, -1)$ satisfies $4 + 3 = 7$ and $4 + 2 + 3 = 9$, so it lies on L.

Reflection in a plane preserves distances and fixes every point of the plane, in particular every point of L. So for every point X of L, $XA_\lambda = XA$.

Let N be the foot of the perpendicular from A to L and $r = AN$. Then A_λ is at distance r from N, and A_λ lies in the plane through A perpendicular to L (the segment AA_λ is perpendicular to Π_λ, hence to L, and its midpoint lies in Π_λ). So every A_λ lies on the circle with centre N and radius r in that perpendicular plane.

Conversely, for a point $Y \neq A$ of this circle, the perpendicular bisector plane of AY contains N and the direction of L, so it is a plane through L in which Y is the image of A. Therefore the images fill the whole circle except the single point that is the image of A in $4x + y - 3z = 9$.

Two images are at most a diameter $2r$ apart, and a diameter not using the missing point is attained by two members of the family. So the maximum distance is $2r$.

$\overrightarrow{P_0A} = (2-1,\ 2-2,\ -2+1) = (1, 0, -1)$, $|\overrightarrow{P_0A}|^2 = 2$; its component along L is $\dfrac{(1, 0, -1) \cdot (3, 0, 4)}{5} = -\dfrac{1}{5}$.

$r^2 = 2 - \dfrac{1}{25} = \dfrac{49}{25}$, so $r = \dfrac{7}{5} = 1.4$ and $2r = 2.8$.

Final Answer: 2.8

Question 39

English

Consider the family of planes $(4x + 4y + 7z - 4) + \lambda(2x + 3y + 2z - 1) = 0$, $\lambda \in \mathbb{R}$. Each plane of this family meets the xy-plane in a line. Let Π be the plane of the family whose line of intersection with the xy-plane is at the greatest distance from the point $(5, 3, 0)$. Then the cosine of the acute angle between Π and the xy-plane is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

समतलों के कुल $(4x + 4y + 7z - 4) + \lambda(2x + 3y + 2z - 1) = 0$, $\lambda \in \mathbb{R}$ पर विचार कीजिए। इस कुल का प्रत्येक तल xy-तल को एक रेखा में काटता है। माना कि Π इस कुल का वह तल है जिसकी xy-तल के साथ प्रतिच्छेदन रेखा बिन्दु $(5, 3, 0)$ से अधिकतम दूरी पर है। तब Π तथा xy-तल के बीच के न्यून कोण की कोज्या (cosine) _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Putting $z = 0$, the plane of the family meets the xy-plane in the line $(4x+4y-4)+\lambda(2x+3y-1) = 0$, i.e. $(4 + 2\lambda)x + (4 + 3\lambda)y - (4 + \lambda) = 0$. (The coefficients of x and y never vanish together, since $4 + 2\lambda = 0$ gives $\lambda = -2$ and then $4 + 3\lambda = -2 \neq 0$; so each member is a genuine line.)

These lines form the family of lines through the point of intersection of $x+y-1 = 0$ and $2x+3y-1 = 0$: subtracting twice the first from the second gives $y = -1$, so $x = 2$ and $M = (2, -1)$.

For any line through M, its distance from $Q(5, 3)$ is $QM\,|\sin\phi| \leq QM$, where ϕ is the angle between the line and MQ. $QM = \sqrt{9 + 16} = 5$, and equality holds only for the line through M perpendicular to $\overrightarrow{MQ} = (3, 4)$, namely $3(x - 2) + 4(y + 1) = 0$, i.e. $3x + 4y - 2 = 0$.

This line belongs to the family when $\dfrac{4 + 2\lambda}{3} = \dfrac{4 + 3\lambda}{4}$, i.e. $16 + 8\lambda = 12 + 9\lambda$, so $\lambda = 4$; then $(4 + 2\lambda,\ 4 + 3\lambda,\ -(4 + \lambda)) = (12, 16, -8) = 4(3, 4, -2)$, as required.

For $\lambda = 4$ the plane is $(4+8)x + (4+12)y + (7+8)z - (4+4) = 0$, i.e. $\Pi :\ 12x + 16y + 15z - 8 = 0$.

The angle between Π and the xy-plane is the angle between their normals $(12, 16, 15)$ and $(0, 0, 1)$: $\cos\theta = \dfrac{15}{\sqrt{144 + 256 + 225}} = \dfrac{15}{\sqrt{625}} = \dfrac{15}{25} = 0.6$.

Final Answer: 0.6

Question 40

English

Let L be the line $\frac{x}{2} = y - 2 = \frac{z-5}{2}$ and let A be the point $(-3, -1, 5)$. Let Π be the plane $x + 2y + \lambda z = k$, where λ and k are real numbers, such that the image of A in Π coincides with the image of A in the line L. Then the value of $\lambda + k$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि L रेखा $\frac{x}{2} = y - 2 = \frac{z-5}{2}$ है तथा A बिन्दु $(-3, -1, 5)$ है। माना कि Π तल $x + 2y + \lambda z = k$ है, जहाँ λ तथा k वास्तविक संख्याएँ हैं, इस प्रकार कि तल Π में A का प्रतिबिम्ब, रेखा L में A के प्रतिबिम्ब के सम्पाती है। तब $\lambda + k$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

A general point of L is $(2t,\ 2+t,\ 5+2t)$ and the direction of L is $(2, 1, 2)$.

For the foot N of the perpendicular from $A(-3, -1, 5)$: $\overrightarrow{AN} = (2t+3,\ t+3,\ 2t)$ and $\overrightarrow{AN} \cdot (2, 1, 2) = 2(2t + 3) + (t + 3) + 2(2t) = 9t + 9 = 0$, so $t = -1$ and $N = (-2, 1, 3)$.

The image of A in L is B with N the midpoint of AB: $B = 2N - A = (-4+3,\ 2+1,\ 6-5) = (-1, 3, 1)$.

The image of A in a plane is B exactly when the plane is perpendicular to AB and passes through the midpoint of AB. Here $\overrightarrow{AB} = (2, 4, -4) = 2(1, 2, -2)$ and the midpoint is $N(-2, 1, 3)$.

The normal $(1, 2, \lambda)$ of Π must be parallel to $(1, 2, -2)$, so $\lambda = -2$.

Then N lies on $x + 2y - 2z = k$: $k = -2 + 2 - 6 = -6$.

Hence $\lambda + k = -2 - 6 = -8$.

Final Answer: -8

Question 41

English

Let $OABC$ be a tetrahedron with $\overrightarrow{OA} = \vec{a}$, $\overrightarrow{OB} = \vec{b}$ and $\overrightarrow{OC} = \vec{c}$, where $|\vec{a}| = 2$, $|\vec{b}| = 3$, $|\vec{c}| = 6$,

$$\vec{a} \times \left(\vec{b} \times \vec{c}\right) = 4\vec{b} - 2\vec{c} \quad \text{and} \quad \vec{b} \times (\vec{c} \times \vec{a}) = 3\vec{a} + 2\vec{c}.$$

Let P, Q and R be the points such that $\overrightarrow{OP} = \vec{a} \times \vec{b}$, $\overrightarrow{OQ} = \vec{b} \times \vec{c}$ and $\overrightarrow{OR} = \vec{c} \times \vec{a}$. Then the volume of the tetrahedron $OPQR$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $OABC$ एक चतुष्फलक है जिसमें $\overrightarrow{OA} = \vec{a}$, $\overrightarrow{OB} = \vec{b}$ तथा $\overrightarrow{OC} = \vec{c}$ हैं, जहाँ $|\vec{a}| = 2$, $|\vec{b}| = 3$, $|\vec{c}| = 6$,

$$\vec{a} \times \left(\vec{b} \times \vec{c}\right) = 4\vec{b} - 2\vec{c}$$

तथा

$$\vec{b} \times (\vec{c} \times \vec{a}) = 3\vec{a} + 2\vec{c}$$

हैं। माना कि P, Q तथा R ऐसे बिन्दु हैं कि $\overrightarrow{OP} = \vec{a} \times \vec{b}$, $\overrightarrow{OQ} = \vec{b} \times \vec{c}$ तथा $\overrightarrow{OR} = \vec{c} \times \vec{a}$ हैं। तब चतुष्फलक $OPQR$ का आयतन _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Since $OABC$ is a tetrahedron, $\vec{a}$, $\vec{b}$, $\vec{c}$ are non-coplanar; in particular any two of them are non-collinear.

Vector triple product: $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\,\vec{b} - (\vec{a} \cdot \vec{b})\,\vec{c}$. Comparing with $4\vec{b} - 2\vec{c}$ (coefficients are unique because $\vec{b}$, $\vec{c}$ are non-collinear): $\vec{a} \cdot \vec{c} = 4$ and $\vec{a} \cdot \vec{b} = 2$.

Similarly $\vec{b} \times (\vec{c} \times \vec{a}) = (\vec{b} \cdot \vec{a})\,\vec{c} - (\vec{b} \cdot \vec{c})\,\vec{a}$. Comparing with $3\vec{a} + 2\vec{c}$: $\vec{a} \cdot \vec{b} = 2$ (consistent) and $\vec{b} \cdot \vec{c} = -3$.

Volume of $OPQR = \dfrac{1}{6}\left|[\vec{a} \times \vec{b} \;\; \vec{b} \times \vec{c} \;\; \vec{c} \times \vec{a}]\right|$.

$(\vec{b} \times \vec{c}) \times (\vec{c} \times \vec{a}) = [\vec{b} \;\; \vec{c} \;\; \vec{a}]\,\vec{c} - [\vec{b} \;\; \vec{c} \;\; \vec{c}]\,\vec{a} = [\vec{a} \;\; \vec{b} \;\; \vec{c}]\,\vec{c}$, so $[\vec{a} \times \vec{b} \;\; \vec{b} \times \vec{c} \;\; \vec{c} \times \vec{a}] = (\vec{a} \times \vec{b}) \cdot \vec{c}\,[\vec{a} \;\; \vec{b} \;\; \vec{c}] = [\vec{a} \;\; \vec{b} \;\; \vec{c}]^2$.

The square of a scalar triple product equals the determinant of the dot products:

$$[\vec{a} \;\; \vec{b} \;\; \vec{c}]^2 = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix} = \begin{vmatrix} 4 & 2 & 4 \\ 2 & 9 & -3 \\ 4 & -3 & 36 \end{vmatrix}$$

$= 4(9 \cdot 36 - 9) - 2(2 \cdot 36 + 12) + 4(-6 - 36) = 4(315) - 2(84) + 4(-42) = 1260 - 168 - 168 = 924.$

Hence the volume of $OPQR$ is $\dfrac{924}{6} = 154$.

Final Answer: 154

Question 42

English

A parallelepiped has O as a vertex, and A, B, C are the three vertices of the parallelepiped joined to O by edges, with $OA = 2$, $OB = 3$ and $OC = 4$. Let D be the vertex such that $OADB$ is a face, and let G be the vertex opposite to O, so that OG is a diagonal of the parallelepiped. If $OG = 6$ and $CD = 4$, then the value of $\cos\angle AOB$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक समान्तर षट्फलक का एक शीर्ष O है, तथा A, B, C इसके वे तीन शीर्ष हैं जो O से कोरों द्वारा जुड़े हैं, जहाँ $OA = 2$, $OB = 3$ तथा $OC = 4$ है। माना कि D वह शीर्ष है जिसके लिए $OADB$ एक फलक है, तथा G, O के सम्मुख शीर्ष है, जिससे OG समान्तर षट्फलक का एक विकर्ण है। यदि $OG = 6$ तथा $CD = 4$ है, तो $\cos\angle AOB$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Take O as origin and $\overrightarrow{OA} = \vec{a}$, $\overrightarrow{OB} = \vec{b}$, $\overrightarrow{OC} = \vec{c}$, with $|\vec{a}| = 2$, $|\vec{b}| = 3$, $|\vec{c}| = 4$.

$OADB$ is a parallelogram, so $\overrightarrow{OD} = \vec{a} + \vec{b}$; the vertex opposite to O is $\overrightarrow{OG} = \vec{a} + \vec{b} + \vec{c}$; and $\overrightarrow{CD} = \overrightarrow{OD} - \overrightarrow{OC} = \vec{a} + \vec{b} - \vec{c}$.

Expanding: $OG^2 = |\vec{a} + \vec{b}|^2 + 2(\vec{a} + \vec{b}) \cdot \vec{c} + |\vec{c}|^2$ and $CD^2 = |\vec{a} + \vec{b}|^2 - 2(\vec{a} + \vec{b}) \cdot \vec{c} + |\vec{c}|^2$.

Adding: $OG^2 + CD^2 = 2|\vec{a} + \vec{b}|^2 + 2|\vec{c}|^2$ (the parallelogram law in the diagonal section $ODGC$).

$36 + 16 = 2|\overrightarrow{a} + \overrightarrow{b}|^2 + 2(16) \Rightarrow |\overrightarrow{a} + \overrightarrow{b}|^2 = 10.$

$|\overrightarrow{a} + \overrightarrow{b}|^2 = |\overrightarrow{a}|^2 + |\overrightarrow{b}|^2 + 2\,\overrightarrow{a} \cdot \overrightarrow{b} = 13 + 2\,\overrightarrow{a} \cdot \overrightarrow{b} = 10 \Rightarrow \overrightarrow{a} \cdot \overrightarrow{b} = -\frac{3}{2}.$

$\cos\angle AOB = \dfrac{\overrightarrow{a} \cdot \overrightarrow{b}}{|\overrightarrow{a}||\overrightarrow{b}|} = \dfrac{-3/2}{6} = -\dfrac{1}{4} = -0.25.$

(Such a parallelepiped exists: for example $\overrightarrow{a} \cdot \overrightarrow{c} = 2$, $\overrightarrow{b} \cdot \overrightarrow{c} = 3$ gives $OG = 6$, $CD = 4$ and a positive dot-product determinant 450.)

Final Answer: -0.25

Question 43

English

A parallelepiped has the origin O as a vertex, and $A(2, 1, -1)$, $B(1, 3, 2)$ and C are the three vertices of the parallelepiped joined to O by edges. The vertex of the parallelepiped opposite to O is $G(\lambda + 3,\ 2,\ 4)$, where λ is a real number. The volume of the parallelepiped is 15, and the vectors $\overrightarrow{OA}$, $\overrightarrow{OB}$, $\overrightarrow{OC}$, in this order, form a right-handed system. Then the value of λ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक समान्तर षट्फलक का एक शीर्ष मूलबिन्दु O है, तथा $A(2, 1, -1)$, $B(1, 3, 2)$ और C इसके वे तीन शीर्ष हैं जो O से कोरों द्वारा जुड़े हैं। O के सम्मुख शीर्ष $G(\lambda + 3,\ 2,\ 4)$ है, जहाँ λ एक वास्तविक संख्या है। समान्तर षट्फलक का आयतन 15 है, तथा सदिश $\overrightarrow{OA}$, $\overrightarrow{OB}$, $\overrightarrow{OC}$ इसी क्रम में एक दक्षिणावर्ती (right-handed) निकाय बनाते हैं। तब λ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For a parallelepiped with coterminous edges $\overrightarrow{OA}$, $\overrightarrow{OB}$, $\overrightarrow{OC}$, the vertex opposite to O is given by $\overrightarrow{OG} = \overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC}$.

So $\overrightarrow{OC} = \overrightarrow{OG} - \overrightarrow{OA} - \overrightarrow{OB} = (\lambda + 3 - 2 - 1,\ 2 - 1 - 3,\ 4 + 1 - 2) = (\lambda, -2, 3)$.

The volume is $\left|[\overrightarrow{OA}\ \overrightarrow{OB}\ \overrightarrow{OC}]\right|$, and the system $\overrightarrow{OA}, \overrightarrow{OB}, \overrightarrow{OC}$ is right-handed exactly when $[\overrightarrow{OA}\ \overrightarrow{OB}\ \overrightarrow{OC}] > 0$.

$$[\overrightarrow{OA}\ \overrightarrow{OB}\ \overrightarrow{OC}] = \begin{vmatrix} 2 & 1 & -1 \\ 1 & 3 & 2 \\ \lambda & -2 & 3 \end{vmatrix} = 2(9 + 4) - 1(3 - 2\lambda) + (-1)(-2 - 3\lambda)$$

$= 26 - 3 + 2\lambda + 2 + 3\lambda = 25 + 5\lambda.$

Volume 15: $|25 + 5\lambda| = 15$, so $25 + 5\lambda = 15$ or $25 + 5\lambda = -15$, i.e. $\lambda = -2$ or $\lambda = -8$.

Right-handed needs $25 + 5\lambda > 0$, so $\lambda = -2$ (then $C = (-2, -2, 3)$ and the box product is 15).

Final Answer: -2

Question 44

English

Let $ABCDEFGH$ be a parallelepiped in which $ABCD$ is a face and AE, BF, CG, DH are edges, where $A = (1, 0, 2)$, $B = (7, 1, 2)$, $D = (2, 7, 1)$ and $E = (2, 2, 10)$. Then the volume of the tetrahedron $ACFH$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $ABCDEFGH$ एक समान्तर षट्फलक है जिसमें $ABCD$ एक फलक है तथा AE, BF, CG, DH इसकी कोरें हैं, जहाँ $A = (1, 0, 2)$, $B = (7, 1, 2)$, $D = (2, 7, 1)$ तथा $E = (2, 2, 10)$ हैं। तब चतुष्फलक $ACFH$ का आयतन _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Coterminous edges at A: $\vec{a} = \overrightarrow{AB} = (6, 1, 0)$, $\vec{b} = \overrightarrow{AD} = (1, 7, -1)$, $\vec{c} = \overrightarrow{AE} = (1, 2, 8)$.

$ABCD$ is a parallelogram, so $C = B + D - A = (8, 8, 1)$. The edges BF and DH are equal and parallel to AE, so $F = B + \overrightarrow{AE} = (8, 3, 10)$ and $H = D + \overrightarrow{AE} = (3, 9, 9)$.

$\overrightarrow{AC} = \vec{a} + \vec{b} = (7, 8, -1)$, $\overrightarrow{AF} = \vec{a} + \vec{c} = (7, 3, 8)$, $\overrightarrow{AH} = \vec{b} + \vec{c} = (2, 9, 7)$.

Volume of tetrahedron $ACFH = \dfrac{1}{6}\left|[\overrightarrow{AC}\ \overrightarrow{AF}\ \overrightarrow{AH}]\right|$, with

$$[\overrightarrow{AC}\ \overrightarrow{AF}\ \overrightarrow{AH}] = \begin{vmatrix} 7 & 8 & -1 \\ 7 & 3 & 8 \\ 2 & 9 & 7 \end{vmatrix} = 7(21 - 72) - 8(49 - 16) + (-1)(63 - 6) = -357 - 264 - 57 = -678.$$

So the volume is $\dfrac{678}{6} = 113$.

Check: $[\vec{a} + \vec{b}\ \ \vec{a} + \vec{c}\ \ \vec{b} + \vec{c}] = -2[\vec{a}\ \ \vec{b}\ \ \vec{c}]$, and $[\vec{a}\ \ \vec{b}\ \ \vec{c}] = 6(56 + 2) - 1(8 + 1) + 0 = 339$, so the tetrahedron has one third of the parallelepiped's volume: $\dfrac{339}{3} = 113$.

Final Answer: 113

Question 45

English

Let $\vec{a}$, $\vec{b}$, $\vec{c}$ be three non-coplanar vectors and let x be a real number with $0 < x < 1$. The volume of the parallelepiped with coterminous edges $\vec{a} + x\vec{b} + x^2\vec{c}$, $\vec{b} + x\vec{c} + x^2\vec{a}$ and $\vec{c} + x\vec{a} + x^2\vec{b}$ is $\frac{49}{64}$ times the volume of the parallelepiped with coterminous edges $\vec{a}$, $\vec{b}$, $\vec{c}$. Then the value of x is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $\vec{a}$, $\vec{b}$, $\vec{c}$ तीन असमतलीय सदिश हैं तथा x एक वास्तविक संख्या है जहाँ $0 < x < 1$ है। सहकोटिक (coterminous) कोरों $\vec{a} + x\vec{b} + x^2\vec{c}$, $\vec{b} + x\vec{c} + x^2\vec{a}$ तथा $\vec{c} + x\vec{a} + x^2\vec{b}$ वाले समान्तर षट्फलक का आयतन, सहकोटिक कोरों $\vec{a}$, $\vec{b}$, $\vec{c}$ वाले समान्तर षट्फलक के आयतन का $\frac{49}{64}$ गुना है। तब x का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $\vec{u} = \vec{a} + x\vec{b} + x^2\vec{c}$, $\vec{v} = x^2\vec{a} + \vec{b} + x\vec{c}$, $\vec{w} = x\vec{a} + x^2\vec{b} + \vec{c}$.

For vectors written in the basis $\vec{a}$, $\vec{b}$, $\vec{c}$, the box product is the determinant of the coefficients times $[\vec{a}\ \vec{b}\ \vec{c}]$:

$$[\vec{u}\ \vec{v}\ \vec{w}] = \Delta\,[\vec{a}\ \vec{b}\ \vec{c}], \qquad \Delta = \begin{vmatrix} 1 & x & x^2 \\ x^2 & 1 & x \\ x & x^2 & 1 \end{vmatrix}$$

Since $[\vec{a}\ \vec{b}\ \vec{c}] \neq 0$, the ratio of the volumes is $|\Delta|$.

Each row sums to $1 + x + x^2$. Applying $C_1 \to C_1 + C_2 + C_3$ makes every entry of the first column $1 + x + x^2$; taking this factor out leaves first column $(1, 1, 1)^T$.

Then $R_2 \to R_2 - R_1$ and $R_3 \to R_3 - R_1$:

$$\Delta = (1 + x + x^2)\begin{vmatrix} 1 & x & x^2 \\ 0 & 1 - x & x - x^2 \\ 0 & x^2 - x & 1 - x^2 \end{vmatrix}$$

$= (1 + x + x^2)\left[(1 - x)(1 - x^2) - (x - x^2)(x^2 - x)\right] = (1 + x + x^2)\left[(1 - x)^2(1 + x) + x^2(1 - x)^2\right]$

$= (1 + x + x^2)^2(1 - x)^2 = (1 - x^3)^2 \geq 0.$

So $(1 - x^3)^2 = \frac{49}{64}$, giving $1 - x^3 = \frac{7}{8}$ or $1 - x^3 = -\frac{7}{8}$.

The second gives $x^3 = \frac{15}{8} > 1$, impossible for $0 < x < 1$. The first gives $x^3 = \frac{1}{8}$, so $x = \frac{1}{2} = 0.5$.

Final Answer: 0.5

Question 46

English

Let α, β, γ be the roots of the equation $x^3 - 3x^2 + 1 = 0$. Let A, B, C be the points whose position vectors with respect to the origin O are $\hat{i} + \alpha\hat{j} + \alpha^3\hat{k}$, $\hat{i} + \beta\hat{j} + \beta^3\hat{k}$ and $\hat{i} + \gamma\hat{j} + \gamma^3\hat{k}$, respectively. Then the volume of the tetrahedron $OABC$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $\alpha,\ \beta,\ \gamma$ समीकरण $x^3 - 3x^2 + 1 = 0$ के मूल हैं। माना कि $A,\ B,\ C$ वे बिन्दु हैं जिनके मूलबिन्दु O के सापेक्ष स्थिति सदिश क्रमशः $\hat{i}+\alpha\hat{j}+\alpha^3\hat{k},\ \hat{i}+\beta\hat{j}+\beta^3\hat{k}$ तथा $\hat{i}+\gamma\hat{j}+\gamma^3\hat{k}$ हैं। तब चतुष्फलक $OABC$ का आयतन _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For a tetrahedron with one vertex at the origin, volume $= \frac{1}{6}\left|\left[\overrightarrow{OA}\ \overrightarrow{OB}\ \overrightarrow{OC}\right]\right|$, and the scalar triple product is the determinant of the components:

$$\left[\overrightarrow{OA}\ \overrightarrow{OB}\ \overrightarrow{OC}\right] = \Delta = \begin{vmatrix} 1 & \alpha & \alpha^3 \\ 1 & \beta & \beta^3 \\ 1 & \gamma & \gamma^3 \end{vmatrix}$$

Subtracting the first row from the second and third rows: $R_2 - R_1$ gives $(0,\ \beta-\alpha,\ \beta^3-\alpha^3)$ and $R_3 - R_1$ gives $(0,\ \gamma-\alpha,\ \gamma^3-\alpha^3)$.

Taking the factors $(\beta-\alpha)$ and $(\gamma-\alpha)$ out of these rows: $\Delta = (\beta-\alpha)(\gamma-\alpha)\begin{vmatrix} 1 & \beta^2+\alpha\beta+\alpha^2 \\ 1 & \gamma^2+\alpha\gamma+\alpha^2 \end{vmatrix}$.

The 2×2 determinant is $\gamma^2-\beta^2+\alpha(\gamma-\beta) = (\gamma-\beta)(\alpha+\beta+\gamma)$.

So $\Delta = (\beta-\alpha)(\gamma-\alpha)(\gamma-\beta)(\alpha+\beta+\gamma)$ and $|\Delta| = |(\alpha-\beta)(\beta-\gamma)(\gamma-\alpha)|\,|\alpha+\beta+\gamma|$.

From $x^3 - 3x^2 + 0\cdot x + 1 = 0$: $\alpha+\beta+\gamma = 3$, $\alpha\beta+\beta\gamma+\gamma\alpha = 0$, $\alpha\beta\gamma = -1$.

For the product of differences write $f(x) = x^3 - 3x^2 + 1 = (x-\alpha)(x-\beta)(x-\gamma)$. Differentiating the product and putting $x = \alpha$ gives $f'(\alpha) = (\alpha-\beta)(\alpha-\gamma)$; similarly $f'(\beta) = (\beta-\alpha)(\beta-\gamma)$ and $f'(\gamma) = (\gamma-\alpha)(\gamma-\beta)$.

Multiplying, $f'(\alpha)f'(\beta)f'(\gamma) = -\left[(\alpha-\beta)(\beta-\gamma)(\gamma-\alpha)\right]^2$ (three sign changes).

Now $f'(x) = 3x^2-6x = 3x(x-2)$, so $f'(\alpha)f'(\beta)f'(\gamma) = 27\,\alpha\beta\gamma\,(\alpha-2)(\beta-2)(\gamma-2)$. Here $\alpha\beta\gamma = -1$ and $(\alpha-2)(\beta-2)(\gamma-2) = -f(2) = -(8-12+1) = 3$, so the product is -81.

Hence $\left[(\alpha-\beta)(\beta-\gamma)(\gamma-\alpha)\right]^2 = 81$ and $|(\alpha-\beta)(\beta-\gamma)(\gamma-\alpha)| = 9$ (the roots are real and distinct).

Hence $|\Delta| = 9\times 3 = 27$ and the volume is $\dfrac{27}{6} = 4.5$.

Final Answer: 4.5

Question 47

English

Let $\overrightarrow{a} = 2\hat{i} - \hat{j} + \hat{k}$ and let $\overrightarrow{r}$ be a vector satisfying $\overrightarrow{r} + \overrightarrow{a} \times (\overrightarrow{a} \times \overrightarrow{r}) = -11\hat{i} - 7\hat{j} + 12\hat{k}$. Then the value of $|\overrightarrow{r} \times \overrightarrow{a}|^2$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $\overrightarrow{a} = 2\hat{i} - \hat{j} + \hat{k}$ है तथा $\overrightarrow{r}$ एक ऐसा सदिश है जो $\overrightarrow{r} + \overrightarrow{a} \times (\overrightarrow{a} \times \overrightarrow{r}) = -11\hat{i} - 7\hat{j} + 12\hat{k}$ को संतुष्ट करता है। तब $|\overrightarrow{r} \times \overrightarrow{a}|^2$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Write $\overrightarrow{c} = -11\hat{i} - 7\hat{j} + 12\hat{k}$. Here $|\overrightarrow{a}|^2 = 4+1+1 = 6$.

By the vector triple product, $\overrightarrow{a} \times (\overrightarrow{a} \times \overrightarrow{r}) = (\overrightarrow{a}\cdot\overrightarrow{r})\overrightarrow{a} - (\overrightarrow{a}\cdot\overrightarrow{a})\overrightarrow{r} = (\overrightarrow{a}\cdot\overrightarrow{r})\overrightarrow{a} - 6\overrightarrow{r}$.

The given equation becomes $\overrightarrow{r} + (\overrightarrow{a}\cdot\overrightarrow{r})\overrightarrow{a} - 6\overrightarrow{r} = \overrightarrow{c}$, that is $(\overrightarrow{a}\cdot\overrightarrow{r})\overrightarrow{a} - 5\overrightarrow{r} = \overrightarrow{c}$.

The unknown scalar $\vec{a}\cdot\vec{r}$ is found by taking the dot product with $\vec{a}$: $(\vec{a}\cdot\vec{r})|\vec{a}|^2 - 5(\vec{a}\cdot\vec{r}) = \vec{a}\cdot\vec{c}$.

$\vec{a}\cdot\vec{c} = 2(-11)+(-1)(-7)+1(12) = -22+7+12 = -3$, so $6(\vec{a}\cdot\vec{r}) - 5(\vec{a}\cdot\vec{r}) = -3$, i.e. $\vec{a}\cdot\vec{r} = -3$.

Then $5\vec{r} = (\vec{a}\cdot\vec{r})\vec{a} - \vec{c} = -3(2\hat{i}-\hat{j}+\hat{k}) + 11\hat{i} + 7\hat{j} - 12\hat{k} = 5\hat{i} + 10\hat{j} - 15\hat{k}$, so $\vec{r} = \hat{i} + 2\hat{j} - 3\hat{k}$ (unique).

Check: $\vec{a}\cdot\vec{r} = 2 - 2 - 3 = -3$, consistent.

By Lagrange's identity, $|\vec{r}\times\vec{a}|^2 = |\vec{r}|^2|\vec{a}|^2 - (\vec{r}\cdot\vec{a})^2 = 14\times 6 - (-3)^2 = 84 - 9 = 75$.

Final Answer: 75

Question 48

English

Let $\vec{a}$, $\vec{b}$, $\vec{c}$, $\vec{d}$ be vectors such that $|\vec{c}| = 1$, $|\vec{d}| = 2$, $\vec{c}\cdot\vec{d} = 1$, $\vec{a}\cdot\vec{c} = 2$, $\vec{a}\cdot\vec{d} = -\frac{1}{2}$, $\vec{b}\cdot\vec{c} = 1$ and $\vec{b}\cdot\vec{d} = 2$. Let $\vec{p} = \vec{a}\times\left(\vec{b}\times\left(\vec{c}\times\vec{d}\right)\right)$. Then the value of $\vec{p}\cdot\left(\vec{c}\times\vec{d}\right)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $\vec{a}$, $\vec{b}$, $\vec{c}$, $\vec{d}$ ऐसे सदिश हैं कि $|\vec{c}| = 1$, $|\vec{d}| = 2$, $\vec{c}\cdot\vec{d} = 1$, $\vec{a}\cdot\vec{c} = 2$, $\vec{a}\cdot\vec{d} = -\frac{1}{2}$, $\vec{b}\cdot\vec{c} = 1$ तथा $\vec{b}\cdot\vec{d} = 2$ है। माना कि $\vec{p} = \vec{a}\times\left(\vec{b}\times\left(\vec{c}\times\vec{d}\right)\right)$ है। तब $\vec{p}\cdot\left(\vec{c}\times\vec{d}\right)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

First expand the inner triple product: $\vec{b}\times(\vec{c}\times\vec{d}) = (\vec{b}\cdot\vec{d})\vec{c} - (\vec{b}\cdot\vec{c})\vec{d} = 2\vec{c} - \vec{d}$.

Then $\vec{p} = \vec{a}\times(2\vec{c}-\vec{d}) = 2(\vec{a}\times\vec{c}) - (\vec{a}\times\vec{d})$.

So $\vec{p}\cdot(\vec{c}\times\vec{d}) = 2(\vec{a}\times\vec{c})\cdot(\vec{c}\times\vec{d}) - (\vec{a}\times\vec{d})\cdot(\vec{c}\times\vec{d})$.

Each term is a scalar product of four vectors: $(\vec{x}\times\vec{y})\cdot(\vec{u}\times\vec{v}) = (\vec{x}\cdot\vec{u})(\vec{y}\cdot\vec{v}) - (\vec{x}\cdot\vec{v})(\vec{y}\cdot\vec{u})$.

With $\vec{x} = \vec{a}$, $\vec{y} = \vec{c}$: $(\vec{a}\cdot\vec{c})(\vec{c}\cdot\vec{d}) - (\vec{a}\cdot\vec{d})(\vec{c}\cdot\vec{c}) = 2\cdot 1 - \left(-\frac{1}{2}\right)\cdot 1 = \frac{5}{2}$.

With $\vec{x} = \vec{a}$, $\vec{y} = \vec{d}$: $(\vec{a}\cdot\vec{c})(\vec{d}\cdot\vec{d}) - (\vec{a}\cdot\vec{d})(\vec{d}\cdot\vec{c}) = 2\cdot 4 - \left(-\frac{1}{2}\right)\cdot 1 = \frac{17}{2}$.

Hence $\vec{p}\cdot(\vec{c}\times\vec{d}) = 2\cdot\frac{5}{2} - \frac{17}{2} = 5 - 8.5 = -3.5$.

The data are consistent: $|\vec{c}\times\vec{d}|^2 = 1\cdot 4 - 1 = 3 \neq 0$, and $\vec{a}$, $\vec{b}$ can be chosen with the given dot products with $\vec{c}$, $\vec{d}$ (any component along $\vec{c}\times\vec{d}$ is free and does not affect the answer).

Final Answer: -3.5

Question 49

English

Let $\vec{a} = \hat{i} + 2\hat{j} + 2\hat{k}$ and $\vec{b} = \hat{i}$. Let x, y, z be the real numbers such that $-7\hat{i} + 10\hat{j} + 34\hat{k} = x\,\vec{a} + y\left(\vec{a} \times \vec{b}\right) + z\left(\vec{a} \times \left(\vec{a} \times \vec{b}\right)\right)$. Then the value of $x^2 + y^2 + z^2$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $\vec{a} = \hat{i} + 2\hat{j} + 2\hat{k}$ तथा $\vec{b} = \hat{i}$ है। माना कि x, y, z वे वास्तविक संख्याएँ हैं जिनके लिए $-7\hat{i} + 10\hat{j} + 34\hat{k} = x\,\vec{a} + y\left(\vec{a} \times \vec{b}\right) + z\left(\vec{a} \times \left(\vec{a} \times \vec{b}\right)\right)$ है। तब $x^2 + y^2 + z^2$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Compute the vectors of the combination: $\vec{a} \times \vec{b} = (\hat{i} + 2\hat{j} + 2\hat{k}) \times \hat{i} = 2\hat{j} - 2\hat{k}$ (since $\hat{j} \times \hat{i} = -\hat{k}$, $\hat{k} \times \hat{i} = \hat{j}$).

$\vec{a} \times (\vec{a} \times \vec{b}) = (\vec{a} \cdot \vec{b})\,\vec{a} - (\vec{a} \cdot \vec{a})\,\vec{b} = 1 \cdot (\hat{i} + 2\hat{j} + 2\hat{k}) - 9\hat{i} = -8\hat{i} + 2\hat{j} + 2\hat{k}$.

$\vec{a}$ is perpendicular to $\vec{a} \times \vec{b}$, and $\vec{a} \times (\vec{a} \times \vec{b})$ is perpendicular to both $\vec{a}$ and $\vec{a} \times \vec{b}$. Being non-zero and mutually perpendicular, the three vectors are non-coplanar, so every vector in space is a unique linear combination of them.

Let $\vec{r} = -7\hat{i} + 10\hat{j} + 34\hat{k}$. Taking the dot product of $\vec{r} = x\,\vec{a} + y(\vec{a} \times \vec{b}) + z(\vec{a} \times (\vec{a} \times \vec{b}))$ with each vector kills the other two terms.

With $\vec{a}$: $\vec{r} \cdot \vec{a} = -7 + 20 + 68 = 81 = 9x \Rightarrow x = 9$.

With $2\hat{j} - 2\hat{k}$: $20 - 68 = -48 = 8y \Rightarrow y = -6$.

With $-8\hat{i} + 2\hat{j} + 2\hat{k}$: $56 + 20 + 68 = 144 = 72z \Rightarrow z = 2$.

Check: $9(1, 2, 2) - 6(0, 2, -2) + 2(-8, 2, 2) = (-7, 10, 34)$.

Therefore $x^2 + y^2 + z^2 = 81 + 36 + 4 = 121$.

Final Answer: 121

Question 50

English

Let $OABC$ be a tetrahedron, and let $\vec{a}$, $\vec{b}$, $\vec{c}$ be the position vectors of A, B, C, respectively, with respect to O. Let D be the point on the side BC with $BD : DC = 2 : 1$, let E be the point on the side CA with $CE : EA = 1 : 2$, and let G be the point of intersection of the lines AD and BE. Let M be the mid-point of OA. The line MG meets the plane OBC at the point P, and $\overrightarrow{OP} = p\,\vec{b} + q\,\vec{c}$. Then the value of $p - q$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $OABC$ एक चतुष्फलक है, तथा $\vec{a}$, $\vec{b}$, $\vec{c}$ क्रमशः A, B, C के O के सापेक्ष स्थिति सदिश हैं। माना कि D भुजा BC पर वह बिन्दु है जिसके लिए $BD : DC = 2 : 1$ है, E भुजा CA पर वह बिन्दु है जिसके लिए $CE : EA = 1 : 2$ है, तथा G रेखाओं AD और BE का प्रतिच्छेदन बिन्दु है। माना कि M, OA का मध्य-बिन्दु है। रेखा MG तल OBC से बिन्दु P पर मिलती है, तथा $\overrightarrow{OP} = p\,\vec{b} + q\,\vec{c}$ है। तब $p - q$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

By the section formula, D divides BC in the ratio $2 : 1$, so $\overrightarrow{OD} = \dfrac{1 \cdot \vec{b} + 2\vec{c}}{3}$; E divides CA in the ratio $1 : 2$, so $\overrightarrow{OE} = \dfrac{2\vec{c} + 1 \cdot \vec{a}}{3}$.

In the plane ABC the lines AD and BE meet at one point G. A point of AD is $\vec{a} + s(\overrightarrow{OD} - \vec{a}) = (1 - s)\vec{a} + \dfrac{s}{3}\vec{b} + \dfrac{2s}{3}\vec{c}$.

A point of BE is $\vec{b} + t(\overrightarrow{OE} - \vec{b}) = \dfrac{t}{3}\vec{a} + (1 - t)\vec{b} + \dfrac{2t}{3}\vec{c}$.

Since $\vec{a}$, $\vec{b}$, $\vec{c}$ are non-coplanar, a vector has only one expression in terms of them, so the coefficients are equal: $\dfrac{2s}{3} = \dfrac{2t}{3}$, $1 - s = \dfrac{t}{3}$, $\dfrac{s}{3} = 1 - t$.

From the first, $s = t$; then $1 - s = \dfrac{s}{3}$ gives $s = \dfrac{3}{4}$, and the third equation also holds.

So $\overrightarrow{OG} = \dfrac{1}{4}\vec{a} + \dfrac{1}{4}\vec{b} + \dfrac{1}{2}\vec{c}$ (the coefficients add to 1, as they must for a point of plane ABC).

$\overrightarrow{OM} = \dfrac{1}{2}\vec{a}$. A point of line MG is $\overrightarrow{OM} + k(\overrightarrow{OG} - \overrightarrow{OM}) = \left(\dfrac{1}{2} - \dfrac{k}{4}\right)\vec{a} + \dfrac{k}{4}\vec{b} + \dfrac{k}{2}\vec{c}$.

A point lies in the plane OBC exactly when it is a combination of $\vec{b}$ and $\vec{c}$ alone, that is when the coefficient of $\vec{a}$ is 0 (again by uniqueness): $\dfrac{1}{2} - \dfrac{k}{4} = 0$, $k = 2$.

Then $\overrightarrow{OP} = \dfrac{1}{2}\vec{b} + \vec{c}$, so $p = \dfrac{1}{2}$, $q = 1$ and $p - q = -\dfrac{1}{2} = -0.5$.

Final Answer: -0.5

Question 51

English

Let λ be a real number, and let $\vec{u} = 2\hat{i} + \hat{j} + \lambda\hat{k}$, $\vec{v} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{w} = 4\lambda\hat{i} + \hat{j} + \hat{k}$. Suppose that the vector $3\hat{i} - \hat{j} - 4\hat{k}$ can be expressed in the form $x\vec{u} + y\vec{v} + z\vec{w}$, with real numbers x, y, z, in more than one way. Then the value of λ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि λ एक वास्तविक संख्या है, तथा $\vec{u} = 2\hat{i} + \hat{j} + \lambda\hat{k}$, $\vec{v} = \hat{i} + \hat{j} + \hat{k}$ और $\vec{w} = 4\lambda\hat{i} + \hat{j} + \hat{k}$ हैं। माना कि सदिश $3\hat{i} - \hat{j} - 4\hat{k}$ को वास्तविक संख्याओं x, y, z के साथ $x\vec{u} + y\vec{v} + z\vec{w}$ के रूप में एक से अधिक प्रकार से व्यक्त किया जा सकता है। तब λ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

If $\vec{u}, \vec{v}, \vec{w}$ are non-coplanar, every vector is a linear combination of them in exactly one way. So more than one expression forces them to be coplanar, i.e. $[\vec{u}\ \vec{v}\ \vec{w}] = 0$.

$$\begin{vmatrix} 2 & 1 & \lambda \\ 1 & 1 & 1 \\ 4\lambda & 1 & 1 \end{vmatrix} = 2(1-1) - 1(1-4\lambda) + \lambda(1-4\lambda) = (4\lambda - 1)(1-\lambda)$$

So $\lambda = 1$ or $\lambda = \dfrac{1}{4}$. Coplanarity is necessary, not sufficient: the given vector must also lie in their plane.

Equating components of $x\vec{u}+y\vec{v}+z\vec{w} = 3\hat{i}-\hat{j}-4\hat{k}$ gives the system $2x+y+4\lambda z = 3$, $x+y+z = -1$, $\lambda x + y + z = -4$.

For $\lambda = 1$ the second and third equations read $x + y + z = -1$ and $x + y + z = -4$, which is inconsistent: no expression at all.

For $\lambda = \dfrac{1}{4}$: the third equation is $\dfrac{1}{4}x + y + z = -4$; subtracting it from the second gives $\dfrac{3}{4}x = 3$, so $x = 4$ and $y + z = -5$.

The first equation becomes $8 + y + z = 3$, i.e. $y + z = -5$, the same condition. So $x = 4$, $y = t$, $z = -5 - t$ for every real t: infinitely many expressions.

Hence $\lambda = \dfrac{1}{4} = 0.25$.

Final Answer: 0.25

Question 52

English

Let $\vec{a} = \hat{i} + \hat{k}$ and let $\vec{r}$ be a vector satisfying $\vec{r} \times (\vec{r} \times \vec{a}) = -5\hat{i} - 7\hat{j} + 2\hat{k}$. Then the value of $|\vec{r}|^2$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $\vec{a} = \hat{i} + \hat{k}$ है तथा $\vec{r}$ एक ऐसा सदिश है जो $\vec{r} \times (\vec{r} \times \vec{a}) = -5\hat{i} - 7\hat{j} + 2\hat{k}$ को संतुष्ट करता है। तब $|\vec{r}|^2$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Write $\vec{c} = -5\hat{i}-7\hat{j}+2\hat{k}$ and put $s = |\vec{r}|^2$, $m = \vec{r}\cdot\vec{a}$. Here $|\vec{a}|^2 = 2$ and $\vec{a}\cdot\vec{c} = -5+0+2 = -3$.
By the vector triple product, $\vec{r}\times(\vec{r}\times\vec{a}) = (\vec{r}\cdot\vec{a})\vec{r}-(\vec{r}\cdot\vec{r})\vec{a}$, so the equation is $m\vec{r}-s\vec{a} = \vec{c}$.
The equation is not linear in $\vec{r}$, so instead of components use dot products.
Dot with $\vec{r}$: $ms - sm = \vec{r}\cdot\vec{c}$, so $\vec{r}\cdot\vec{c} = 0$ (as expected, since $\vec{r}\times(\cdots)$ is perpendicular to $\vec{r}$).
Dot with $\vec{c}$: $|\vec{c}|^2 = m(\vec{r}\cdot\vec{c}) - s(\vec{a}\cdot\vec{c}) = 0 - s(-3) = 3s$.
$|\vec{c}|^2 = 25 + 49 + 4 = 78$, hence $s = 26$.

Consistency: dot with $\vec{a}$ gives $m^2 - 2s = -3$, so $m^2 = 49$, $m = \pm 7$; then $\vec{r} = \dfrac{\vec{c}+s\vec{a}}{m} = \dfrac{21\hat{i}-7\hat{j}+28\hat{k}}{\pm 7} = \pm(3\hat{i}-\hat{j}+4\hat{k})$, and both satisfy the equation with $|\vec{r}|^2 = 26$.

Final Answer: 26

Question 53

English

Let $A(1,2,-3)$, $B(-5,-4,0)$ and $C(3,-4,0)$ be the vertices of a triangle ABC. In the plane of the triangle, let $J(\alpha,\beta,\gamma)$ be the point of intersection of the internal bisector of $\angle BAC$ with the external bisectors of $\angle ABC$ and $\angle ACB$. Then the value of $\alpha+\beta+\gamma$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $A(1,2,-3)$, $B(-5,-4,0)$ तथा $C(3,-4,0)$ त्रिभुज ABC के शीर्ष हैं। त्रिभुज के तल में, माना कि $J(\alpha,\beta,\gamma)$ कोण $\angle BAC$ के आन्तरिक समद्विभाजक तथा कोणों $\angle ABC$ और $\angle ACB$ के बाह्य समद्विभाजकों का प्रतिच्छेदन बिन्दु है। तब $\alpha+\beta+\gamma$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The internal bisector of $\angle A$ and the external bisectors of $\angle B$ and $\angle C$ meet at the excentre I_1 opposite A, whose position vector is $\dfrac{-a\vec{A}+b\vec{B}+c\vec{C}}{-a+b+c}$, where $a = BC$, $b = CA$, $c = AB$.
(Reason: the internal bisector from A meets BC at D with $BD : DC = c : b$; the external bisector from B divides AD externally in the ratio $AB : BD$, which gives these weights.)
Side lengths: $\vec{BC} = (8,0,0)$, $a = 8$; $\vec{CA} = (-2,6,-3)$, $b = \sqrt{4+36+9} = 7$; $\vec{AB} = (-6,-6,3)$, $c = \sqrt{36+36+9} = 9$.
$-a+b+c = -8+7+9 = 8$.
$-8\vec{A} = (-8,-16,24)$, $7\vec{B} = (-35,-28,0)$, $9\vec{C} = (27,-36,0)$; the sum is $(-16,-80,24)$.
So $J = \left(\dfrac{-16}{8}, \dfrac{-80}{8}, \dfrac{24}{8}\right) = (-2,-10,3)$.
Check: J is at distance $3\sqrt{5}$ from each of the lines AB, BC, CA, and lies on the other side of BC from A.
$\alpha+\beta+\gamma = -2-10+3 = -9$.

Final Answer: -9

Question 54

English

Let $P(1,-2,3)$, $Q(-5,-5,9)$ and $R(5,0,7)$ be the vertices of a triangle PQR. The internal bisector of $\angle QPR$ meets the side QR at the point D. Then the length of the segment PD is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $P(1,-2,3)$, $Q(-5,-5,9)$ तथा $R(5,0,7)$ त्रिभुज PQR के शीर्ष हैं। कोण $\angle QPR$ का आन्तरिक समद्विभाजक भुजा QR से बिन्दु D पर मिलता है। तब रेखाखण्ड PD की लम्बाई _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$\overrightarrow{PQ} = Q - P = (-6,-3,6)$, so $PQ = \sqrt{36+9+36} = 9$; $\overrightarrow{PR} = R - P = (4,2,4)$, so $PR = \sqrt{16+4+16} = 6$.

The internal bisector of $\angle QPR$ divides QR internally in the ratio of the adjacent sides: $QD : DR = PQ : PR = 9 : 6 = 3 : 2$.

(In vector form, $\overrightarrow{PD}$ is along $\dfrac{\overrightarrow{PQ}}{9} + \dfrac{\overrightarrow{PR}}{6}$, which gives the same point.)

By the section formula, $D = \dfrac{2Q + 3R}{3+2} = \dfrac{2(-5,-5,9) + 3(5,0,7)}{5} = \dfrac{(5,-10,39)}{5} = \left(1,-2,\dfrac{39}{5}\right)$.

$\overrightarrow{PD} = \left(0,0,\dfrac{39}{5} - 3\right) = \left(0,0,\dfrac{24}{5}\right)$, so $PD = \dfrac{24}{5} = 4.8$.

Check: $\cos\angle QPD = \dfrac{\overrightarrow{PQ} \cdot \overrightarrow{PD}}{9 \cdot 4.8} = \dfrac{6 \cdot 4.8}{43.2} = \dfrac{2}{3}$ and $\cos\angle RPD = \dfrac{4 \cdot 4.8}{6 \cdot 4.8} = \dfrac{2}{3}$, equal as required.

Final Answer: 4.8

Question 55

English

A ray of light starts from the point $B(5,1)$ in the xy-plane, travels along a straight line and strikes a plane mirror, which is perpendicular to the xy-plane, at the point $A(2,-3)$. After reflection the ray passes through the point $C(-3,9)$. The line in which the mirror meets the xy-plane meets the y-axis at the point $(0,k)$. Then the value of k is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

प्रकाश की एक किरण xy-तल में बिन्दु $B(5,1)$ से चलकर एक सरल रेखा के अनुदिश जाती है तथा एक समतल दर्पण से, जो xy-तल के लम्बवत् है, बिन्दु $A(2,-3)$ पर टकराती है। परावर्तन के पश्चात किरण बिन्दु $C(-3,9)$ से होकर जाती है। जिस रेखा में दर्पण xy-तल को काटता है, वह y-अक्ष से बिन्दु $(0,k)$ पर मिलती है। तब k का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

At the point of incidence the incident ray AB (traced back) and the reflected ray AC make equal angles with the normal to the mirror, on opposite sides. So the normal at A bisects $\angle BAC$ internally, and the mirror line is the other (external) bisector of the angle between the lines AB and AC.

Vector form of the internal bisector: its direction is $\hat{u}+\hat{v}$, the sum of the unit vectors along $\overrightarrow{AB}$ and $\overrightarrow{AC}$.

$\overrightarrow{AB} = 3\hat{i}+4\hat{j}$, $AB=5$, $\hat{u}=\dfrac{3\hat{i}+4\hat{j}}{5}$; $\overrightarrow{AC} = -5\hat{i}+12\hat{j}$, $AC=13$, $\hat{v}=\dfrac{-5\hat{i}+12\hat{j}}{13}$.

$\hat{u}+\hat{v}=\dfrac{39\hat{i}+52\hat{j}-25\hat{i}+60\hat{j}}{65}=\dfrac{14\hat{i}+112\hat{j}}{65}$, parallel to $\hat{i}+8\hat{j}$.

The mirror line passes through $A(2,-3)$ and is perpendicular to $\hat{i}+8\hat{j}$: $1\cdot(x-2)+8(y+3)=0$, that is $x+8y+22=0$.

The same result from the angle bisectors of two lines: AB is $4x-3y-17=0$ and AC is $12x+5y-9=0$; their bisectors are $\dfrac{4x-3y-17}{5}=\pm\dfrac{12x+5y-9}{13}$.

The $+$ sign gives $52x-39y-221=60x+25y-45$, i.e. $x+8y+22=0$; the $-$ sign gives $112x-14y-266=0$, i.e. $8x-y-19=0$.

The line $8x-y-19=0$ has direction $\hat{i}+8\hat{j}$, so it is the internal bisector of $\angle BAC$ (the normal); the mirror is $x+8y+22=0$.

Check by reflection: the incident direction $\dfrac{-3\hat{i}-4\hat{j}}{5}$ reflected in $x+8y+22=0$ becomes $\dfrac{-5\hat{i}+12\hat{j}}{13}$, the direction of $\overrightarrow{AC}$.

Putting $x=0$: $8y=-22$, $k=-\dfrac{11}{4}=-2.75$.

Final Answer: -2.75

Question 56

English

Let ABC be a triangle with $BC = 4$, $CA = 5$ and $AB = 6$. As P varies over all points in space, the minimum value of $PA^2 + PB^2 + 2PC^2$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि ABC एक त्रिभुज है जिसमें $BC = 4$, $CA = 5$ तथा $AB = 6$ है। जब P अन्तरिक्ष के सभी बिन्दुओं पर परिवर्तित होता है, तब $PA^2 + PB^2 + 2PC^2$ का न्यूनतम मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Choose the origin at the point G whose position vector is $\dfrac{\vec{A} + \vec{B} + 2\vec{C}}{4}$; then the position vectors $\vec{a}, \vec{b}, \vec{c}$ of A, B, C satisfy $\vec{a} + \vec{b} + 2\vec{c} = \vec{0}$. Let P have position vector $\vec{p}$.

$PA^2 = |\vec{p} - \vec{a}|^2 = |\vec{p}|^2 - 2\vec{p} \cdot \vec{a} + |\vec{a}|^2$, and similarly for PB^2, PC^2.

$S = PA^2 + PB^2 + 2PC^2 = 4|\vec{p}|^2 - 2\vec{p} \cdot (\vec{a} + \vec{b} + 2\vec{c}) + |\vec{a}|^2 + |\vec{b}|^2 + 2|\vec{c}|^2 = 4|\vec{p}|^2 + K.$

Hence $S \geq K$ with equality only for $\vec{p} = \vec{0}$, i.e. $P = G$ (a point of the plane ABC). The minimum is $K = |\vec{a}|^2 + |\vec{b}|^2 + 2|\vec{c}|^2$.

To express K by the sides, expand: $AB^2 = |\vec{a} - \vec{b}|^2$, $BC^2 = |\vec{b} - \vec{c}|^2$, $CA^2 = |\vec{c} - \vec{a}|^2$, so

$AB^2 + 2BC^2 + 2CA^2 = 3|\vec{a}|^2 + 3|\vec{b}|^2 + 4|\vec{c}|^2 - 2\vec{a} \cdot \vec{b} - 4\vec{b} \cdot \vec{c} - 4\vec{c} \cdot \vec{a}.$

Also $0 = |\vec{a} + \vec{b} + 2\vec{c}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + 4|\vec{c}|^2 + 2\vec{a} \cdot \vec{b} + 4\vec{b} \cdot \vec{c} + 4\vec{c} \cdot \vec{a}.$

Adding the two lines, the dot products cancel: $AB^2 + 2BC^2 + 2CA^2 = 4|\vec{a}|^2 + 4|\vec{b}|^2 + 8|\vec{c}|^2 = 4K.$

So $K = \dfrac{AB^2 + 2BC^2 + 2CA^2}{4} = \dfrac{36 + 2 \cdot 16 + 2 \cdot 25}{4} = \dfrac{118}{4} = 29.5.$

Final Answer: 29.5

Question 57

English

Let A, B, C, D be four non-coplanar points such that $AB = 3$, $BC = 5$, $CD = 9$, $DA = 7$ and $AC = 4$, and the acute angle between the lines AC and BD is $\cos^{-1}\left(\frac{1}{3}\right)$. Let M and N be the mid-points of the segments AC and BD, respectively. Then the value of MN^2 is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि A, B, C, D चार असमतलीय बिन्दु इस प्रकार हैं कि $AB = 3$, $BC = 5$, $CD = 9$, $DA = 7$ तथा $AC = 4$ है, तथा रेखाओं AC और BD के बीच का न्यून कोण $\cos^{-1}\left(\frac{1}{3}\right)$ है। माना कि M तथा N क्रमशः रेखाखण्डों AC तथा BD के मध्य-बिन्दु हैं। तब MN^2 का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Take position vectors $\vec{a}$, $\vec{b}$, $\vec{c}$, $\vec{d}$ of A, B, C, D. Each squared length is a dot product, for example $AB^2 = |\vec{b} - \vec{a}|^2$.

$AB^2+CD^2-BC^2-DA^2 = |\vec{b}-\vec{a}|^2+|\vec{d}-\vec{c}|^2-|\vec{c}-\vec{b}|^2-|\vec{a}-\vec{d}|^2$. The squares $|\vec{a}|^2, \ldots, |\vec{d}|^2$ cancel and the result is $2(\vec{b}\cdot\vec{c}+\vec{a}\cdot\vec{d}-\vec{a}\cdot\vec{b}-\vec{c}\cdot\vec{d}) = 2(\vec{b}-\vec{d})\cdot(\vec{c}-\vec{a}) = 2\,\overrightarrow{DB}\cdot\overrightarrow{AC}$.

Hence $\overrightarrow{DB}\cdot\overrightarrow{AC} = \dfrac{3^2+9^2-5^2-7^2}{2} = \dfrac{16}{2} = 8$.

If ϕ is the acute angle between the lines, $|\overrightarrow{AC}\cdot\overrightarrow{DB}| = AC\cdot BD\cos\phi$, so $8 = 4\cdot BD\cdot\dfrac{1}{3}$ and $BD = 6$.

Next, $\overrightarrow{MN} = \dfrac{\vec{b}+\vec{d}}{2} - \dfrac{\vec{a}+\vec{c}}{2} = \dfrac{(\vec{b}-\vec{a})+(\vec{d}-\vec{c})}{2}$.

$4MN^2 = AB^2 + CD^2 + 2(\vec{b}-\vec{a})\cdot(\vec{d}-\vec{c})$, and expanding as before $2(\vec{b}-\vec{a})\cdot(\vec{d}-\vec{c}) = BC^2 + DA^2 - BD^2 - AC^2$.

So $4MN^2 = AB^2 + BC^2 + CD^2 + DA^2 - AC^2 - BD^2 = 9 + 25 + 81 + 49 - 16 - 36 = 112$.

Therefore $MN^2 = 28$. (The six lengths satisfy all triangle inequalities and give a tetrahedron of positive volume, so the configuration exists.)

Final Answer: 28

Question 58

English

For $\theta, \phi \in \left(0, \frac{\pi}{2}\right)$, let $\overrightarrow{u}(\theta, \phi)$ denote the unit vector which makes an angle θ with the xy-plane, has a positive z-component, and whose projection on the xy-plane makes an angle ϕ with the positive x-axis. Let $\theta_1, \theta_2, \phi_1, \phi_2 \in \left(0, \frac{\pi}{2}\right)$ be such that $\tan\theta_1 = \frac{3}{4}$, $\tan\theta_2 = \frac{12}{5}$, $\tan\phi_1 = \frac{4}{3}$ and $\tan\phi_2 = \frac{7}{24}$. If ψ is the angle between the vectors $\overrightarrow{u}(\theta_1, \phi_1)$ and $\overrightarrow{u}(\theta_2, \phi_2)$, then the value of $\cos\psi$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$\theta, \phi \in \left(0, \frac{\pi}{2}\right)$ के लिए, माना कि $\overrightarrow{u}(\theta, \phi)$ वह मात्रक सदिश है जो xy-तल से θ कोण बनाता है, जिसका z-घटक धनात्मक है, तथा xy-तल पर जिसका प्रक्षेप धनात्मक x-अक्ष से ϕ कोण बनाता है। माना कि $\theta_1, \theta_2, \phi_1, \phi_2 \in \left(0, \frac{\pi}{2}\right)$ इस प्रकार हैं कि $\tan\theta_1 = \frac{3}{4}$, $\tan\theta_2 = \frac{12}{5}$, $\tan\phi_1 = \frac{4}{3}$ तथा $\tan\phi_2 = \frac{7}{24}$ है। यदि सदिशों $\overrightarrow{u}(\theta_1, \phi_1)$ तथा $\overrightarrow{u}(\theta_2, \phi_2)$ के बीच का कोण ψ है, तो $\cos\psi$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

A unit vector making angle θ with the xy-plane has z-component $\sin\theta$ (positive here) and a projection of length $\cos\theta$ on the xy-plane. That projection makes angle ϕ with the x-axis, so $\overrightarrow{u}(\theta, \phi) = \cos\theta\cos\phi\,\hat{i} + \cos\theta\sin\phi\,\hat{j} + \sin\theta\,\hat{k}$; its length is $\sqrt{\cos^2\theta + \sin^2\theta} = 1$.

For unit vectors, $\cos\psi = \overrightarrow{u}(\theta_1, \phi_1) \cdot \overrightarrow{u}(\theta_2, \phi_2) = \cos\theta_1\cos\theta_2\cos\phi_1\cos\phi_2 + \cos\theta_1\cos\theta_2\sin\phi_1\sin\phi_2 + \sin\theta_1\sin\theta_2$.

By the compound-angle formula $\cos\phi_1\cos\phi_2 + \sin\phi_1\sin\phi_2 = \cos(\phi_1 - \phi_2)$, so $\cos\psi = \cos\theta_1\cos\theta_2\cos(\phi_1 - \phi_2) + \sin\theta_1\sin\theta_2$.

All angles are acute, so all ratios are positive: $\tan\theta_1 = \frac{3}{4} \Rightarrow \cos\theta_1 = \frac{4}{5},\ \sin\theta_1 = \frac{3}{5}$; $\tan\theta_2 = \frac{12}{5} \Rightarrow \cos\theta_2 = \frac{5}{13},\ \sin\theta_2 = \frac{12}{13}$.

$\tan\phi_1 = \frac{4}{3} \Rightarrow \cos\phi_1 = \frac{3}{5},\ \sin\phi_1 = \frac{4}{5}$; $\tan\phi_2 = \frac{7}{24} \Rightarrow \cos\phi_2 = \frac{24}{25},\ \sin\phi_2 = \frac{7}{25}$.

$\cos(\phi_1 - \phi_2) = \frac{3}{5} \cdot \frac{24}{25} + \frac{4}{5} \cdot \frac{7}{25} = \frac{72+28}{125} = \frac{4}{5}$.

$\cos\psi = \frac{4}{5} \cdot \frac{5}{13} \cdot \frac{4}{5} + \frac{3}{5} \cdot \frac{12}{13} = \frac{16}{65} + \frac{36}{65} = \frac{52}{65} = \frac{4}{5} = 0.8$.

Final Answer: 0.8

Question 59

English

Let $OABC$ be a tetrahedron in which the edge OA is perpendicular to the edge BC and the edge OB is perpendicular to the edge CA. Suppose $OA = 6$, $OB = 8$, $OC = 10$ and $\angle AOB = \frac{\pi}{3}$. Then the value of $AB^2 + BC^2 + CA^2$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $OABC$ एक चतुष्फलक है जिसमें कोर OA कोर BC पर लम्ब है तथा कोर OB कोर CA पर लम्ब है। माना कि $OA = 6$, $OB = 8$, $OC = 10$ तथा $\angle AOB = \frac{\pi}{3}$ है। तब $AB^2 + BC^2 + CA^2$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Take O as the origin and $\overrightarrow{OA} = \vec{a}$, $\overrightarrow{OB} = \vec{b}$, $\overrightarrow{OC} = \vec{c}$, with $|\vec{a}| = 6$, $|\vec{b}| = 8$, $|\vec{c}| = 10$.

From $\angle AOB = \frac{\pi}{3}$: $\vec{a} \cdot \vec{b} = 6 \cdot 8 \cdot \frac{1}{2} = 24$.

$OA \perp BC$ means $\vec{a} \cdot (\vec{c} - \vec{b}) = 0$, so $\vec{a} \cdot \vec{c} = \vec{a} \cdot \vec{b} = 24$.

$OB \perp CA$ means $\vec{b} \cdot (\vec{a} - \vec{c}) = 0$, so $\vec{b} \cdot \vec{c} = \vec{a} \cdot \vec{b} = 24$.

(Then $\vec{c} \cdot (\vec{b} - \vec{a}) = 24 - 24 = 0$, which proves that the third pair OC, AB is also perpendicular.)

So $\cos\angle BOC = \dfrac{\vec{b} \cdot \vec{c}}{8 \cdot 10} = \dfrac{3}{10}$ and $\cos\angle COA = \dfrac{\vec{c} \cdot \vec{a}}{10 \cdot 6} = \dfrac{2}{5}$.

Cosine rule in triangle AOB: $AB^2 = 6^2 + 8^2 - 2 \cdot 6 \cdot 8 \cdot \frac{1}{2} = 100 - 48 = 52$.

Cosine rule in triangle BOC: $BC^2 = 8^2 + 10^2 - 2 \cdot 8 \cdot 10 \cdot \frac{3}{10} = 164 - 48 = 116$.

Cosine rule in triangle COA: $CA^2 = 10^2 + 6^2 - 2 \cdot 10 \cdot 6 \cdot \frac{2}{5} = 136 - 48 = 88$.

$AB^2 + BC^2 + CA^2 = 52 + 116 + 88 = 256$. (The Gram determinant of $\vec{a}$, $\vec{b}$, $\vec{c}$ is $142848 > 0$, so such a tetrahedron exists.)

Final Answer: 256

Question 60

English

Let $A(6,-2,6)$, $B(-4,-2,1)$, $C(-4,3,-3)$ and $D(-3,-4,5)$ be four points in space. Let P be the point on the line AB which is equidistant from C and D. If $\overrightarrow{AP} = \lambda\overrightarrow{PB}$, then the value of λ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $A(6,-2,6)$, $B(-4,-2,1)$, $C(-4,3,-3)$ तथा $D(-3,-4,5)$ आकाश में चार बिन्दु हैं। माना कि P रेखा AB पर वह बिन्दु है जो C तथा D से समदूरस्थ है। यदि $\overrightarrow{AP} = \lambda\overrightarrow{PB}$ है, तो λ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Every point of the line AB is $P = A+t(B-A)$ with $B-A = (-10, 0, -5)$, so $P = (6-10t,\ -2,\ 6-5t)$.
$P - C = (10 - 10t,\ -5,\ 9 - 5t)$ and $P - D = (9 - 10t,\ 2,\ 1 - 5t)$.
By the distance formula, $PC^2 = (10-10t)^2+25+(9-5t)^2 = 100-200t+100t^2+25+81-90t+25t^2 = 206 - 290t + 125t^2$.
$PD^2 = (9 - 10t)^2 + 4 + (1 - 5t)^2 = 81 - 180t + 100t^2 + 4 + 1 - 10t + 25t^2 = 86 - 190t + 125t^2$.
$PC^2 = PD^2$: the quadratic terms cancel and $206 - 290t = 86 - 190t$, so $t = \dfrac{120}{100} = \dfrac{6}{5}$. Hence $P = (-6, -2, 0)$, a unique point.
$\overrightarrow{AP} = \lambda\overrightarrow{PB}$ means $P = \dfrac{A + \lambda B}{1 + \lambda}$ (section formula), that is $P = A + \dfrac{\lambda}{1 + \lambda}(B - A)$, so $\dfrac{\lambda}{1 + \lambda} = \dfrac{6}{5}$, giving $5\lambda = 6 + 6\lambda$ and $\lambda = -6$.
Check: $\overrightarrow{AP} = (-12, 0, -6)$ and $\overrightarrow{PB} = (2, 0, 1)$, so $\overrightarrow{AP} = -6\,\overrightarrow{PB}$; P divides AB externally, beyond B. Also $PC^2 = 4 + 25 + 9 = 38 = 9 + 4 + 25 = PD^2$.

Final Answer: -6

Question 61

English

Let $A(1,-1,-2)$, $B(3,1,-4)$, $C(-2,4,-1)$ and $D(0,-3,-1)$ be the vertices of a tetrahedron. Let M be the mid-point of the edge AB, let N be the point on the edge AC such that $AN : NC = 1 : 2$, and let G be the centroid of the triangle BCD. The plane through M, N and D meets the line AG at the point $X(\alpha, \beta, \gamma)$. Then the value of $\alpha + \beta + \gamma$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $A(1,-1,-2)$, $B(3,1,-4)$, $C(-2,4,-1)$ तथा $D(0,-3,-1)$ एक चतुष्फलक के शीर्ष हैं। माना कि M कोर AB का मध्य-बिन्दु है, N कोर AC पर वह बिन्दु है जिसके लिए $AN : NC = 1 : 2$ है, तथा G त्रिभुज BCD का केन्द्रक है। बिन्दुओं M, N तथा D से होकर जाने वाला तल रेखा AG से बिन्दु $X(\alpha, \beta, \gamma)$ पर मिलता है। तब $\alpha + \beta + \gamma$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Mid-point formula: $M = \left(\frac{1+3}{2}, \frac{-1+1}{2}, \frac{-2-4}{2}\right) = (2, 0, -3)$.

Section formula with $AN : NC = 1 : 2$: $N = \frac{2A + C}{3} = \left(\frac{2-2}{3}, \frac{-2+4}{3}, \frac{-4-1}{3}\right) = \left(0, \frac{2}{3}, -\frac{5}{3}\right)$.

Centroid: $G = \frac{B + C + D}{3} = \left(\frac{1}{3}, \frac{2}{3}, -2\right)$.

Normal of the plane MND: $\overrightarrow{MN} = \left(-2, \frac{2}{3}, \frac{4}{3}\right)$, $\overrightarrow{MD} = (-2, -3, 2)$, and $\overrightarrow{MN} \times \overrightarrow{MD} = \left(\frac{2}{3} \cdot 2 - \frac{4}{3}(-3),\ \frac{4}{3}(-2) - \right.$ $\left(\frac{16}{3}, \frac{4}{3}, \frac{22}{3}\right)$, parallel to $(8, 2, 11)$.

Plane through $M(2, 0, -3)$: $8(x-2) + 2y + 11(z+3) = 0$, that is $8x + 2y + 11z + 17 = 0$. Check: N gives $0 + \frac{4}{3} - \frac{55}{3} + 17 = 0$ and D gives $0 - 6 - 11 + 17 = 0$.

$G - A = \left(-\frac{2}{3}, \frac{5}{3}, 0\right)$, so the line AG is $\left(1 - \frac{2t}{3},\ -1 + \frac{5t}{3},\ -2\right)$, $t \in \mathbb{R}$.

Substituting in the plane: $8 - \frac{16t}{3} - 2 + \frac{10t}{3} - 22 + 17 = 1 - 2t = 0$, so $t = \frac{1}{2}$ (the plane meets AG at its mid-point).

$X = \left(1 - \frac{1}{3}, -1 + \frac{5}{6}, -2\right) = \left(\frac{2}{3}, -\frac{1}{6}, -2\right)$.

$\alpha + \beta + \gamma = \frac{2}{3} - \frac{1}{6} - 2 = \frac{4 - 1 - 12}{6} = -\frac{9}{6} = -1.5$.

Final Answer: -1.5

Question 62

English

Let $L_1 : \dfrac{x}{1} = \dfrac{y-2}{4} = \dfrac{z+1}{1}$ and $L_2 : \dfrac{x+1}{3} = \dfrac{y-2}{2} = \dfrac{z-6}{-2}$. Let T be the line through the origin O that intersects both L_1 and L_2, and let T meet L_1 at the point A and L_2 at the point B. If d is the shortest distance between L_1 and L_2, then the value of $\dfrac{AB}{d}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $L_1 : \dfrac{x}{1} = \dfrac{y-2}{4} = \dfrac{z+1}{1}$ तथा $L_2 : \dfrac{x+1}{3} = \dfrac{y-2}{2} = \dfrac{z-6}{-2}$ हैं। माना कि T मूलबिन्दु O से होकर जाने वाली वह रेखा है जो L_1 तथा L_2 दोनों को प्रतिच्छेद करती है, तथा T रेखा L_1 से बिन्दु A पर और L_2 से बिन्दु B पर मिलती है। यदि L_1 तथा L_2 के बीच की न्यूनतम दूरी d है, तो $\dfrac{AB}{d}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Write $\vec{u}_1 = (1, 4, 1)$ through $P_1(0, 2, -1)$ for L_1, and $\vec{u}_2 = (3, 2, -2)$ through $P_2(-1, 2, 6)$ for L_2.

Let T pass through O with direction (l, m, n). Two non-parallel lines intersect if and only if they are coplanar, that is, the determinant of the vector joining a point of each line and the two directions is zero.

T and L_1: $\begin{vmatrix} 0 & 2 & -1 \\ 1 & 4 & 1 \\ l & m & n \end{vmatrix} = 0(4n - m) - 2(n - l) - 1(m - 4l) = 6l - m - 2n = 0.$

T and L_2: $\begin{vmatrix} -1 & 2 & 6 \\ 3 & 2 & -2 \\ l & m & n \end{vmatrix} = -1(2n + 2m) - 2(3n + 2l) + 6(3m - 2l) = -16l + 16m - 8n = 0$, that is $2l - 2m + n = 0$.

From $6l - m - 2n = 0$ and $2l - 2m + n = 0$: adding twice the second to the first gives $10l - 5m = 0$, so $m = 2l$ and then $n = 2m - 2l = 2l$. Hence $l : m : n = 1 : 2 : 2$ and T is $x = k,\ y = 2k,\ z = 2k$.

Meeting L_1: $(s,\ 2 + 4s,\ -1 + s) = (k, 2k, 2k)$ gives $s = k$ and $2 + 4k = 2k$, so $k = -1$; check z: $-1 + (-1) = -2 = 2k$. So $A = (-1, -2, -2)$.

Meeting L_2: $(-1 + 3t,\ 2 + 2t,\ 6 - 2t) = (k, 2k, 2k)$ gives $2 + 2t = 2(-1 + 3t)$, so $t = 1$, $k = 2$; check z: $6 - 2 = 4 = 2k$. So $B = (2, 4, 4)$.

$AB = |(3, 6, 6)| = 9$. (Both points lie on T, on opposite sides of O.)

Shortest distance: $\vec{u}_1 \times \vec{u}_2 = (4(-2) - 1 \cdot 2,\ 1 \cdot 3 - 1 \cdot (-2),\ 1 \cdot 2 - 4 \cdot 3) = (-10, 5, -10)$, of magnitude 15.

$\overrightarrow{P_1P_2} = (-1, 0, 7)$, so $d = \dfrac{|(-1, 0, 7) \cdot (-10, 5, -10)|}{15} = \dfrac{|10 - 70|}{15} = 4$; since $d \neq 0$ the lines are skew, so T is unique.

$\dfrac{AB}{d} = \dfrac{9}{4} = 2.25$ (as expected, $AB \geq d$, since A and B lie on the two lines).

Final Answer: 2.25

Question 63

English

Let L_1 be the line $\vec{r} = (5\hat{i}-\hat{j}+3\hat{k})+s(\hat{i}-\hat{k})$, $s \in \mathbb{R}$, and let L_2 be the line $\vec{r} = (3\hat{i}+5\hat{j}+5\hat{k})+t(2\hat{j}-\hat{k})$, $t \in \mathbb{R}$. As P varies on L_1 and Q varies on L_2, the locus of the mid-point of the segment PQ is a plane which meets the x-axis, the y-axis and the z-axis at $(a, 0, 0)$, $(0, b, 0)$ and $(0, 0, c)$, respectively. Then the value of $a + b + c$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि L_1 रेखा $\vec{r} = (5\hat{i} - \hat{j} + 3\hat{k}) + s(\hat{i} - \hat{k})$, $s \in \mathbb{R}$ है, तथा L_2 रेखा $\vec{r} = (3\hat{i} + 5\hat{j} + 5\hat{k}) + t(2\hat{j} - \hat{k})$, $t \in \mathbb{R}$ है। जब P रेखा L_1 पर तथा Q रेखा L_2 पर परिवर्तित होते हैं, तब रेखाखण्ड PQ के मध्य-बिन्दु का बिन्दुपथ एक तल है जो x-अक्ष, y-अक्ष तथा z-अक्ष से क्रमशः $(a, 0, 0)$, $(0, b, 0)$ तथा $(0, 0, c)$ पर मिलता है। तब $a + b + c$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

General points: $P = (5 + s, -1, 3 - s)$ on L_1 and $Q = (3, 5 + 2t, 5 - t)$ on L_2.

The mid-point (h, k, l) of PQ has $h = \dfrac{8+s}{2} = 4+\dfrac{s}{2}$, $k = \dfrac{4+2t}{2} = 2+t$, $l = \dfrac{8-s-t}{2} = 4-\dfrac{s}{2}-\dfrac{t}{2}$.

From the first two, $s = 2h - 8$ and $t = k - 2$. Substituting in the third: $l = 4 - (h - 4) - \dfrac{k-2}{2}$, so $2l = 16 - 2h - k + 2$, that is $2h + k + 2l = 18$.

Conversely, for any (h, k) the values $s = 2h - 8$, $t = k - 2$ give a mid-point with that h, k and with l from the plane equation, so every point of the plane is attained. The locus is the plane $2x + y + 2z = 18$.

(This plane passes through the mid-point $(4, 2, 4)$ of the given points and is parallel to both lines; its normal is $(1, 0, -1) \times (0, 2, -1) = (2, 1, 2)$. The lines are skew because $(-2, 6, 2) \cdot (2, 1, 2) = 6 \neq 0$.)

Intercepts: $y = z = 0$ gives $a = 9$; $x = z = 0$ gives $b = 18$; $x = y = 0$ gives $c = 9$.

$a + b + c = 9 + 18 + 9 = 36$.

Final Answer: 36

Question 64

English

Let A be the point $(3, -2, 4)$. A point P moves in space so that the sum of the squares of its distances from the three coordinate axes is equal to $3PA^2$. The locus of P is a sphere of radius r. Then the value of r^2 is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि A बिन्दु $(3, -2, 4)$ है। एक बिन्दु P आकाश में इस प्रकार गति करता है कि तीनों निर्देशांक अक्षों से इसकी दूरियों के वर्गों का योग $3PA^2$ के बराबर है। P का बिन्दुपथ त्रिज्या r का एक गोला है। तब r^2 का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For $P(x, y, z)$ the foot of the perpendicular on the x-axis is $(x, 0, 0)$, so the squared distance from the x-axis is $y^2 + z^2$; similarly $z^2 + x^2$ and $x^2 + y^2$ for the y- and z-axes.

Sum of the squares $= 2(x^2 + y^2 + z^2)$.

$PA^2 = (x-3)^2 + (y+2)^2 + (z-4)^2 = x^2 + y^2 + z^2 - 6x + 4y - 8z + 29$.

The condition $2(x^2+y^2+z^2) = 3PA^2$ gives $2(x^2+y^2+z^2) = 3(x^2+y^2+z^2) - 18x + 12y - 24z + 87$, that is $x^2 + y^2 + z^2 - 18x + 12y - 24z + 87 = 0$.

Completing squares: $(x-9)^2 + (y+6)^2 + (z-12)^2 = 81 + 36 + 144 - 87 = 174 > 0$, a sphere with centre $(9, -6, 12)$, that is $3A$.

Hence $r^2 = 174$.

Final Answer: 174

Question 65

English

Let L_1 be the line through the point $(2, 1, 2)$ parallel to the vector $\hat{i} + 2\hat{j} + 2\hat{k}$, and let L_2 be the line through the point $(0, -1, -2)$ parallel to the vector $\hat{i} + 2\hat{j} - 2\hat{k}$. The locus of a point of the xy-plane which is equidistant from L_1 and L_2 meets the x-axis only at $(\alpha, 0, 0)$ and the y-axis only at $(0, \beta, 0)$. Then the value of $\alpha + \beta$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि L_1 बिन्दु $(2, 1, 2)$ से होकर जाने वाली तथा सदिश $\hat{i} + 2\hat{j} + 2\hat{k}$ के समान्तर रेखा है, तथा L_2 बिन्दु $(0, -1, -2)$ से होकर जाने वाली तथा सदिश $\hat{i} + 2\hat{j} - 2\hat{k}$ के समान्तर रेखा है। xy-तल के उस बिन्दु का बिन्दुपथ, जो L_1 तथा L_2 से समदूरस्थ है, x-अक्ष से केवल $(\alpha, 0, 0)$ पर तथा y-अक्ष से केवल $(0, \beta, 0)$ पर मिलता है। तब $\alpha + \beta$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For a point P and a line through A with direction $\vec{u}$, the foot of the perpendicular splits $\overrightarrow{AP}$ into a part along $\vec{u}$ of length $\dfrac{|\overrightarrow{AP} \cdot \vec{u}|}{|\vec{u}|}$ and a perpendicular part, so the distance d satisfies $d^2 = |\overrightarrow{AP}|^2 - \dfrac{(\overrightarrow{AP} \cdot \vec{u})^2}{|\vec{u}|^2}$.

Let $P(x, y, 0)$. For L_1: $A(2, 1, 2)$, $\vec{u} = (1, 2, 2)$, $|\vec{u}|^2 = 9$, $\overrightarrow{AP} = (x - 2, y - 1, -2)$, $\overrightarrow{AP} \cdot \vec{u} = x - 2 + 2y - 2 - 4 = x + 2y - 8$.

$d_1^2 = (x - 2)^2 + (y - 1)^2 + 4 - \dfrac{(x + 2y - 8)^2}{9}$.

For L_2: $B(0, -1, -2)$, $\vec{v} = (1, 2, -2)$, $|\vec{v}|^2 = 9$, $\overrightarrow{BP} = (x, y+1, 2)$, $\overrightarrow{BP} \cdot \vec{v} = x+2y+2-4 = x+2y-2$.

$d_2^2 = x^2 + (y + 1)^2 + 4 - \dfrac{(x + 2y - 2)^2}{9}$.

$d_1^2 - d_2^2 = \left[(x - 2)^2 - x^2\right] + \left[(y - 1)^2 - (y + 1)^2\right] - \dfrac{(x + 2y - 8)^2 - (x + 2y - 2)^2}{9}$.

$(x-2)^2 - x^2 = 4 - 4x$, $(y-1)^2 - (y+1)^2 = -4y$, and $(x+2y-8)^2 - (x+2y-2)^2 = (-6)(2x+4y-10)$.

So $d_1^2 - d_2^2 = 4 - 4x - 4y + \dfrac{6(2x + 4y - 10)}{9} = \dfrac{12 - 12x - 12y + 4x + 8y - 20}{3} = \dfrac{-8x - 4y - 8}{3}$.

Equidistance ($d_1 = d_2$, equivalently $d_1^2 = d_2^2$) gives $2x + y + 2 = 0$: in the xy-plane the locus is this straight line. The x^2, y^2, xy terms cancelled because the directions differ only in the sign of the z-component.

It meets the x-axis only at $x = -1$ and the y-axis only at $y = -2$, so $\alpha = -1$, $\beta = -2$ and $\alpha + \beta = -3$.

Final Answer: -3

Question 66

English

Let S be the sphere $x^2+y^2+z^2-4x+2y-6z-86=0$ and let Π be the plane $x+2y+2z+12=0$. The plane Π cuts S in a circle Γ. The line of intersection of Π with the plane $2x+y-2z=15$ meets Γ at the points P and Q. Let R be any point of the larger of the two arcs of Γ determined by P and Q. Then the value of $\cos\angle PRQ$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि S गोला $x^2+y^2+z^2-4x+2y-6z-86=0$ है तथा Π तल $x+2y+2z+12=0$ है। तल Π गोले S को एक वृत्त Γ में काटता है। तल Π तथा तल $2x+y-2z=15$ की प्रतिच्छेदन रेखा Γ से बिन्दुओं P तथा Q पर मिलती है। माना कि R, P तथा Q द्वारा निर्धारित Γ के दो चापों में से बड़े चाप का कोई बिन्दु है। तब $\cos\angle PRQ$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

S: $(x-2)^2+(y+1)^2+(z-3)^2=4+1+9+86=100$, so the centre is $C(2,-1,3)$ and the radius is 10.

Distance of C from Π: $\dfrac{|2-2+6+12|}{\sqrt{1+4+4}}=\dfrac{18}{3}=6<10$, so Γ is a real circle of radius $\sqrt{100-36}=8$.

Its centre is the foot of the perpendicular from C: $F=C-\dfrac{18}{9}(1,2,2)=(0,-5,-1)$; check $0-10-2+12=0$.

Let Σ be the plane $2x+y-2z=15$. Since $(1,2,2)\cdot(2,1,-2)=2+2-4=0$, $\Sigma\perp\Pi$. For a point F of Π, the perpendicular from F to Σ then lies in Π and meets Σ on the line $\ell=\Pi\cap\Sigma$, so the distance of F from ℓ equals its distance from Σ: $h=\dfrac{|0-5+2-15|}{3}=\dfrac{18}{3}=6$.

$h=6<8$, so ℓ cuts Γ at two points P, Q. The perpendicular FM from the centre to the chord bisects it, and in the right triangle FMP, $FM=6$, $FP=8$, so $\angle PFM=\alpha$ with $\cos\alpha=\dfrac{6}{8}=\dfrac{3}{4}$ and $\angle PFQ=2\alpha$.

$\alpha<\frac{\pi}{2}$, so $\angle PFQ<\pi$ and the larger arc lies on the same side of PQ as F. The angle subtended by a chord at a point of the major arc is half the angle at the centre, so $\angle PRQ=\alpha$ for every such R.

Hence $\cos\angle PRQ=\cos\alpha=\dfrac{3}{4}=0.75$.

Final Answer: 0.75